

Directional coherence and effect magnitude in single-cell CRISPR perturbation responses

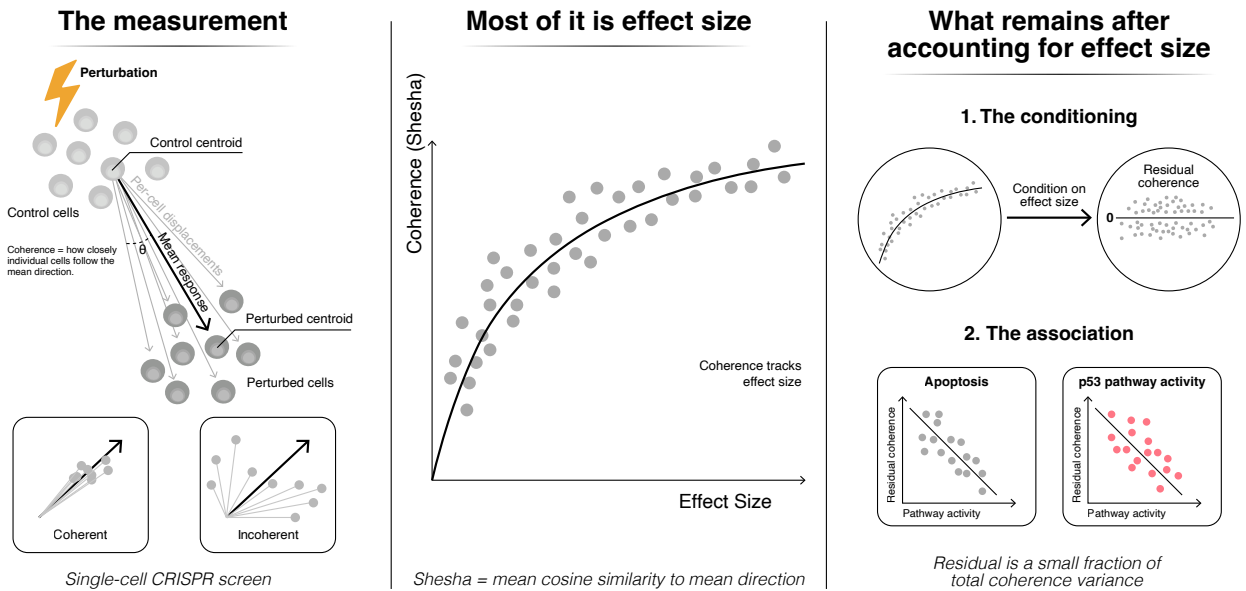
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Abstract

Single-cell CRISPR screens summarize each perturbation by how far cells move from their unperturbed state, averaging over cell-to-cell variation. Whether the cells moved together is not captured: two perturbations with identical effect magnitude can differ qualitatively, one driving cells along a shared trajectory, the other scattering them around the same mean. We define Shesha_p coherence (S_p), the mean cosine similarity between individual cell displacement vectors and their mean, and ask what it adds across six Perturb-seq datasets (2,285 perturbations; CRISPRa, CRISPRi, Cas9 knockout). Coherence and effect magnitude are closely coupled (Spearman $\rho = 0.84$ – 0.98), and the coupling persists in a foundation-model embedding, under alternative effect-size definitions, and when analysis is restricted to responding cells. Associations reported without conditioning on effect size largely recover it, shown here for two comparisons that vanish under conditioning. Effect size accounts for 89–96% of coherence variance; within the remainder, coherence is negatively associated with apoptosis and p53 signalling in the two largest screens. Two standard responder classifiers agree on only 68% of cells. Directional coherence is largely a restatement of effect magnitude, and heterogeneity measures should report their redundancy with effect size as a matter of course. S_p is implemented in the open-source [shesha-geometry](#) Python package.

Graphical Abstract



1 Introduction

Two cells carrying the same genetic perturbation do not respond the same way. In pooled CRISPR screens with single-cell readout, cells assigned an identical guide routinely diverge: some shift far from their unperturbed state and others barely move, some engage one downstream program and others another, and in developmental systems a shared perturbation can leave one cell differentiating and its neighbor stem-like (Replogle et al. 2022; Weinreb et al. 2020). That variation reflects where each cell began, which programs were available to it, and how completely the perturbation took hold. Collapsing it into a mean is what single-cell readout was meant to avoid.

The analysis of these screens has not kept pace with that resolution. A perturbation’s effect is still usually reported as the distance between mean perturbed and mean control expression profiles. Two perturbations with the same summary can differ in kind: one may drive every cell along a shared trajectory, the other may scatter cells in many directions around the same mean. A hit that averages over subpopulations doing different things will not behave like a uniform response, whether the next step is mechanistic follow-up, a predictive model, or a therapeutic target.

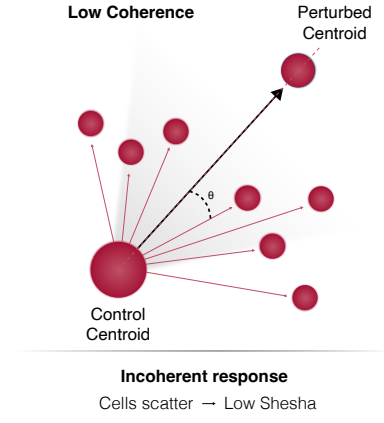
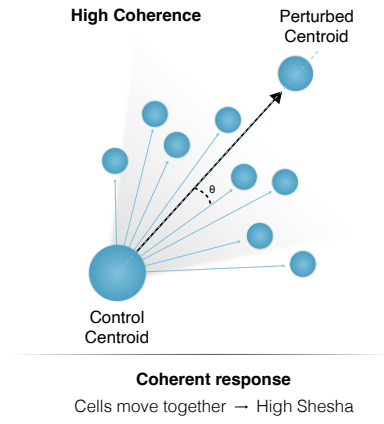
Recent work has begun to recover what the mean discards. Nadig et al. 2025 decompose perturbation effects into shared and gene-specific components through transcriptome-wide differential expression, characterizing how broadly a perturbation acts. Song et al. 2025 estimate a per-cell perturbation-response score, identifying which cells within an assigned guide population responded strongly. Harmonized benchmarking resources (Peidli et al. 2024a) have made comparison across screens possible. None of these asks whether the cells that responded moved in the same direction.

That question is geometric. For each perturbed cell, compute its displacement from the control centroid; for the perturbation, compute the mean of those displacements; then take the average cosine similarity between the individual displacements and their mean. We call this Sheshap coherence (S_p), applying to perturbation biology a measure of geometric self-consistency developed for representational stability more generally (Raju 2026a,c,d). High values indicate cells moving together, low values cells scattering, and the two configurations are in principle separable from how far the cells moved (Fig. 1). Coherence does bound magnitude from below, since aligned displacements produce a longer mean than scattered ones. But that constraint runs one way and says nothing about the joint distribution real perturbations occupy: cells moving a short distance in near-unison, and cells moving far in many directions, are both geometrically admissible and both biologically interpretable.

Separable in principle is not the same as separable in the data. A statistic can be well defined, cheap to compute, and still mostly recover a quantity already in routine use. We treat coherence as a quantity to characterize rather than a metric to promote: how much of it is effect magnitude, what remains after that is accounted for, and what the remainder is associated with. To our knowledge this has not been asked of a perturbation-heterogeneity measure. Answering it takes more than reporting one correlation. It requires conditioning against effect-size measures that do not share the metric’s construction, testing whether the result depends on which cells count as responding, and checking whether rankings survive a change of representation.

We address it in six single-cell CRISPR datasets (Adamson et al. 2016; Dixit et al. 2016; Norman et al. 2019; Papalexi et al. 2021; Replogle et al. 2022) spanning activation, interference and Cas9 knockout in three cell types, 2,285 perturbations under a single frozen configuration. Coherence and effect magnitude are closely coupled (Spearman $\rho = 0.843$ to 0.977). The coupling survives recomputing both quantities in a transformer embedding (scGPT; Cui et al. 2024), redefining effect size as an energy distance or as transcriptome-wide impact from differential expression, and restricting analysis to cells classified as responding to their assigned guide. Associations reported without conditioning on effect size largely recover effect size, including two comparisons that are strongly positive until that step and vanish after it. What remains is a residual of four to eleven percent of coherence variance, within which coherence is negatively associated with apoptosis and p53 signalling in the two largest screens. Those associations are small, concentrated in the largest dataset, and conditional on which family of effect-size measure defines the residual.

a Perturbation Coherence



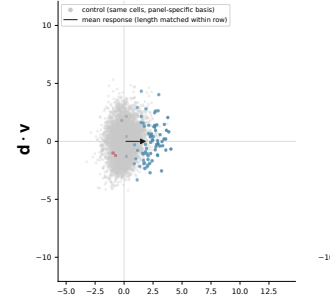
Shesha = mean cosine similarity
to the mean response direction

b Displacement-frame Shesha · Replogle 2022 (CRISPRi), K562

Extreme · Δ Shesha = 0.246

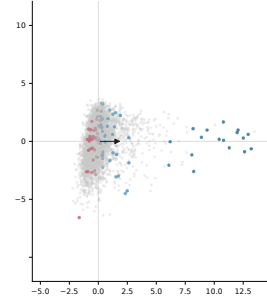
WDR5 · Higher Shesha

Shesha = 0.442 ($\approx$ 94th pct) · $|\mu| = 2.16$



CHMP2A* · Lower Shesha

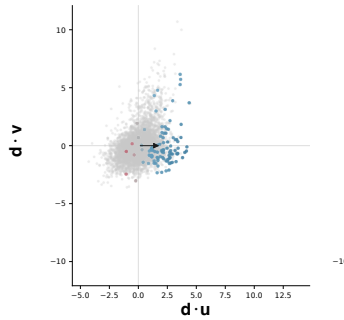
Shesha = 0.197 ($\approx$ 54th pct) · $|\mu| = 2.07$



Typical difference · Δ Shesha = 0.034

UBA2 · Higher Shesha

Shesha = 0.422 ($\approx$ 92th pct) · $|\mu| = 2.06$



RNF20 · Lower Shesha

Shesha = 0.388 ($\approx$ 89th pct) · $|\mu| = 1.96$

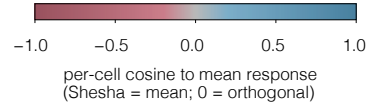
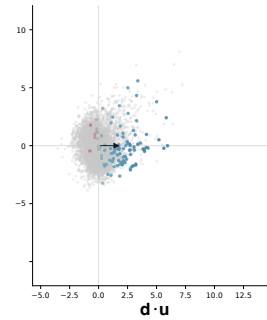


Figure 1: Directional coherence of single-cell perturbation responses. (a) Shesha_p (S_p) is the mean cosine similarity between each perturbed cell's displacement from the control centroid and the mean response direction. Both schematics share the mean response vector; only the angular spread of individual displacements differs, so coherence and effect magnitude are in principle separable. (b) The same contrast in Replogle et al. 2022 (CRISPRi, K562 Replogle et al. 2022). Cells are plotted in each perturbation's own displacement frame, with the horizontal axis along the mean response direction $\mathbf{u}$ and the vertical axis the leading principal component of the residual displacements $\mathbf{v}$; perturbed cells are colored by their individual cosine to the mean response, the quantity S_p averages. Control cells (grey) are the same cells in all four panels, projected into each panel's own basis. Black arrows mark the mean response, matched in length within each row. Upper row shows the pair with the largest coherence difference available at matched effect magnitude; lower row shows the median difference, where the two distributions are visually indistinguishable. Percentiles are within-dataset. Each panel has its own basis, so vertical positions are not comparable across panels, and the two-dimensional projection compresses 49 orthogonal dimensions into one, so visible angles understate the true angular spread. *Partial penetrance on the $\mathbf{u}$ projection (a near-control cell mass alongside a responding tail); this is a per-perturbation observation and responder filtering does not remove the coherence–magnitude relationship in any dataset (Fig. 4).

2 Materials and Methods

2.1 Datasets

Six single-cell CRISPR perturbation datasets were analyzed (**SI Appendix, Table S1**): Norman et al. 2019 (CRISPRa, K562, $n = 236$ perturbations including 105 single-gene and 131 combinatorial) (Norman et al. 2019); Adamson et al. 2016 UPR (CRISPRi, K562, $n = 87$) and Adamson et al. 2016 pilot (CRISPRi, K562, $n = 8$; secondary / underpowered for pathway claims) (Adamson et al. 2016); Dixit et al. 2016 (CRISPR-KO, BMDCs, $n = 98$) (Dixit et al. 2016); Papalexi et al. 2021 (CRISPR-KO, THP-1, $n = 24$) (Papalexi et al. 2021); and Replogle et al. 2022 (genome-scale CRISPRi, K562, $n = 1,832$ perturbations after filtering for ≥ 50 cells per perturbation) (Replogle et al. 2022). Together these comprise $n = 2,285$ perturbations in the frozen Sp table; pathway and stress partials use the five scoreable datasets ($n = 2,277$) after excluding the Adamson pilot. All datasets were accessed via pertpy (v1.0.4) (Heumos, Ji, et al. 2025); Papalexi was additionally rebuilt from GEO (GSE153056) as an independent check that scores are not pertpy-loader-dependent (**SI Appendix, Table S22**).

2.2 Preprocessing

Each dataset was preprocessed independently to prevent cross-dataset batch effects. All manuscript-facing Sp, pathway, stress, QC, and embedding tables use one frozen preprocessing and scoring pipeline (seed 320; version stamps and software pins in **SI Appendix, Tables S2 and S4**). The pipeline consisted of quality filtering (cells with fewer than 100 detected genes removed). Per-dataset matrix scale was pinned from inspection of .X (maximum and fraction near-integer): Norman and Dixit are already log-transformed (`matrix_is_log=True`; normalization and `log1p` skipped), whereas Adamson UPR/pilot, Papalexi, and Replogle are raw counts (`matrix_is_log=False`; **SI Appendix, Table S3**). Raw-count matrices were library-size normalized to the median library size (`target_sum=None`) and `log1p`-transformed. The frozen 50-dimensional linear embedding used two recorded computational backends. Adamson UPR, Adamson pilot, and Papalexi used `scanpy.pp.highly_variable_genes` (2,000 genes) followed by `scanpy.pp.pca`. Norman, Dixit, and Replogle used a memory-safe sparse backend: genes were ranked by variance divided by mean on all retained cells for Dixit or on a seed-320 sample of 20,000 cells for Norman and Replogle, followed by `sklearn.decomposition.TruncatedSVD` (`random_state=320`, `n_iter=5`). Perturbations with fewer than 50 cells were excluded. Large perturbations and controls were downsampled in an order-invariant manner by hashing `seed|obs_name` (at most 100 cells per perturbation and 5,000 controls), so cell selection does not depend on loader row order. All downstream coherence and magnitude computations were performed on this frozen linear embedding. Robustness to embedding dimensionality (10, 20, 30, 50, 100 dimensions), refitting the same dataset-specific backend at each dimension, is confirmed in **SI Appendix, Fig. S3 and Table S7**.

2.3 Control group identification

Control group assignment used a multi-stage matching protocol to accommodate heterogeneous labeling conventions across datasets. Labels were first matched case-insensitively against known control terms (“control”, “ctrl”, “non-targeting”, “NT”, “unperturbed”), then by delimiter-aware regex for short tokens, and finally by substring matching for embedded keywords. Dataset-specific handling: Replogle labels containing “non-targeting” or beginning with “chr” were assigned to control, and “pos_control” labels were removed; Papalexi non-targeting guides (NTg1–NTg7) were pooled into a single control group (2,386 cells); Norman cells labeled “control” served as the control group. Full details are provided in **SI Appendix, Control Group Identification**.

2.4 Shesha perturbation coherence

For each perturbation p applied to n_p cells, we computed the shift vector for each perturbed cell as the difference between its frozen linear-embedding coordinates and the control centroid. The mean perturbation direction is the average of these shift vectors. Shesha_p coherence (S_p) is defined as the mean cosine similarity between individual cell shift vectors and this mean direction:

$$S_p = \frac{1}{n_p} \sum_{i=1}^{n_p} \frac{\mathbf{d}_i \cdot \bar{\mathbf{d}}}{\|\mathbf{d}_i\| \|\bar{\mathbf{d}}\|} \quad (1)$$

where $\mathbf{d}_i = \mathbf{x}_i - \boldsymbol{\mu}_{\text{ctrl}}$ is the shift vector for cell i and $\bar{\mathbf{d}} = \frac{1}{n_p} \sum_i \mathbf{d}_i$ is the mean perturbation direction. A perturbation with

S_p near 1 drives all cells in the same direction (high coherence); S_p near 0 indicates scattering across expression space. Effect magnitude (M_p) is the Euclidean norm of the mean shift vector, $M_p = \|\bar{\mathbf{d}}\|$.

The metric adapts geometric self-consistency from the general Shesha framework (Raju 2026a,c) to perturbation biology. To avoid implying temporal or dynamical stability, we use “coherence” for the measured quantity; S_p remains the mathematical symbol. The authoritative frozen manuscript scores are computed by `pipeline_core.calculate_sp`, the local implementation of Eq. (1). Every manuscript table and figure reporting the primary frozen S_p uses those scores. The pinned `shesha.bio` package is used only by auxiliary callers and does not replace the frozen score table; versions are recorded in **SI Appendix, Reproducibility**. Robustness to distance metric choice (Euclidean, Mahalanobis, k -NN) is evaluated across all six datasets in **SI Appendix, Fig. S2 and Table S6**. The uniform frozen filter excludes perturbations represented by fewer than 50 cells. The Adamson pilot ($n = 8$ perturbations) is retained in the six-dataset S_p table but excluded from pathway and stress partials.

2.5 E-distance competitor

To test whether the S_p -magnitude relationship is an artifact of centroid shift, we computed the `scPerturb / pertpy` energy distance (Peidli et al. 2024a) from each perturbation to the control distribution in the same frozen 50-PC embedding and the same hash-selected cells used for S_p . E-distance is the Euclidean U -statistic $2\mathbb{E}\|X - Y\| - \mathbb{E}_{i \neq j}\|X_i - X_j\| - \mathbb{E}_{i \neq j}\|Y_i - Y_j\|$, using off-diagonal within-group means. The signed estimate is retained (it can be negative when the two distributions are close) and is never labelled magnitude; centroid magnitude M_p remains the primary size measure. The full frozen control set (at most 5,000 cells) is used for every perturbation. Per dataset we report Spearman $\rho(M_p, \text{E-distance})$, $\rho(S_p, M_p)$, $\rho(S_p, \text{E-distance})$, and the fraction of rank- S_p variance remaining after rank-OLS residualization on each size measure. Pathway and stress associations were re-fit under three size models (`|centroid`, `|E-distance`, `|centroid+E-distance`) and three QC models (`|centroid+QC`, `|E-distance+QC`, `joint`). Primary survivors are the intersection of those three QC models, not their union. Per-dataset values are in **SI Appendix, Tables S15 and S16**.

2.6 TRADE transcriptome-wide impact

As a non-geometric size competitor we estimated transcriptome-wide impact (TWI) with TRADE (Nadig et al. 2025). TWI is the mixture variance $\text{Var}(\beta)$ of true gene-level $\log_2$ fold-changes from an `ashr` half-uniform mixture with a uniform prior (Stephens 2016). Inputs are Welch $\log_2\text{FC}$ and standard errors versus the frozen control cells on the full log-normalized transcriptome (same hash-selected cells as S_p ; no highly variable gene subset and no PCA). `scanpy log1p` is natural log and is converted to $\log_2$ by dividing by $\ln 2$. Genes with non-finite SE, $|\log_2\text{FC}| > 10$, or matching the perturbed target are excluded. TWI values were computed via a Python implementation of the `ashr` half-uniform mixture variance, numerically consistent with R `ashr` (Stephens 2016). Per dataset we report $\rho(S_p, \text{TWI})$, $\rho(M_p, \text{TWI})$, and the geometric comparators $\rho(S_p, M_p)$ and $\rho(S_p, \text{E-distance})$. TRADE was added after the frozen pipeline lock and is a robustness column: it is not folded into the three-model QC intersection. TWI-conditioned pathway partials are interpreted only after a diagnostic Spearman of TWI against each Hallmark score (centroid magnitude as comparator), because TWI is estimated from gene-level differential expression and Hallmark scores are themselves expression-derived. The Adamson pilot ($n = 8$) is excluded from pathway claims. Details are in **SI Appendix, Tables S16 and S17**.

2.7 Perturbation-response score proxies

Mean per-cell Euclidean and Mahalanobis distances from the control centroid were evaluated as geometric proxies for Song et al.’s perturbation-response score (PS) (Song et al. 2025). Their weak correlations with the `scMAGeCK`-derived score (computed via a faithful Python reimplement of the `scMAGeCK` constrained-optimisation algorithm; the official R package is also supported and yields numerically equivalent results; $\rho = 0.097$ and 0.149 , respectively; **SI Appendix, Table S28**) show that neither distance is a substitute for PS. They are retained only as a proxy caution, not as alternate implementations of PS. The responder-filter sensitivity below uses Song PS as the primary caller and `Mixscape` as a secondary caller.

2.8 Perturbation-efficiency sensitivity

To test whether the magnitude-coherence relationship is an artifact of unperturbed cells within assigned guide pools, we recomputed S_p after retaining only putative responder cells identified by Song et al.’s perturbation-response score (PS);

primary arm; computed via the Python reimplementation of scMAGeCK described above) or Mixscape (Papalexi et al. 2021) (secondary arm on Norman). Responder calling and Sp rescoring use the same frozen preprocess path as the manuscript Sp table; this sensitivity analysis does not replace the frozen Sp table. For PS, cells were labeled responders when within-perturbation rescaled PS ≥ 0.5 . Primary endpoints are the change in Sp-magnitude Spearman ρ after filtering and the rank correlation between Sp on all cells and Sp on responders. Absolute ΔS_p after filtering is not interpreted as recovered coherence, because both callers select along the mean signature direction. The Adamson pilot was excluded by the same thin- n floor used elsewhere ($n < 30$); Papalexi ($n = 24$) is reported in Table S18. Full dataset-level results, PS-Mixscape per-cell agreement, and version stamps are reported in SI Appendix, Tables S19 and S20.

2.9 Split-half directional-agreement assay

We used split-half directional agreement as a descriptive check on sampling precision, not as experimental reproducibility. For each perturbation in Replogle ($n = 1,832$) and Norman ($n = 236$), cells were randomly divided into equal halves. We computed each half's mean shift relative to the same control centroid and recorded the cosine between the two half-shift vectors, repeating this procedure for 50 random splits. Because this endpoint and S_p are derived from the same cell-level shifts and control reference, no predictive or biological-replication claim is made from their association.

The non-circular check used independent guides targeting the same gene. After conditioning on magnitude, gene-level S_p did not predict between-guide directional agreement in Replogle ($n = 104$ genes; $\rho = 0.006$, 95% CI $-0.220, 0.225$) or Papalexi ($n = 23$ genes; $\rho = 0.183$, 95% CI $-0.317, 0.644$; SI Appendix, Table S21). These null results are reported as the relevant test of the stronger reproducibility claim.

2.10 Pathway-level stress scoring

To assess the association between Sheshap coherence and cellular stress, we computed composite pathway scores for five MSigDB Hallmark gene sets (Liberzon et al. 2015): Unfolded Protein Response (UPR), p53 Pathway, Apoptosis, Reactive Oxygen Species (ROS), and mTORC1 Signaling. Pathway scores were computed using `scanpy.tl.score_genes` with curated gene lists (72–78 overlapping genes for UPR, 32–51 for p53, 53–73 for Apoptosis, 45–55 for ROS per dataset) on the full normalized transcriptome before highly variable gene selection, then correlated with perturbation coherence controlling for effect magnitude. Both raw Spearman correlations and partial correlations (controlling for magnitude) are reported, with Benjamini-Hochberg correction applied across the pathway-dataset combinations.

Per-perturbation cell-quality covariates are mean mitochondrial percentage, detected-gene count, and library depth (`percent_mito`, `n_genes`, `n_counts`) over the same frozen cells. When $n < 40$, only mitochondrial percentage is used to avoid rank deficiency. Primary pathway and stress survivors are the intersection of `|centroid+QC`, `|E-distance+QC`, and the joint model. TWI-conditioned partials are a separate robustness column and are not added to that intersection.

Magnitude-conditioned and QC-conditioned associations were adopted when bootstrap confidence intervals excluded 0, $|\rho| > 0.1$, and BH-FDR < 0.05 . Benjamini-Hochberg FDR is computed once per dataset $\times$ covariate model across the nine scored features (five Hallmark pathway scores and four stress markers); the same family is used for `|magnitude`, `|centroid+QC`, `|E-distance+QC`, and the joint model. After observing knife-edge CI bounds and small- n rank deficiency, three post hoc conservative rules were applied: (i) demote otherwise-surviving rows whose nearer $|CI|$ bound is strictly $< \varepsilon = 0.02$ to indeterminate; (ii) use mito-only QC covariates when $n < 40$; (iii) report point estimates only / exclude from the gate when $n < 30$. These post hoc rules and the adopted $\varepsilon = 0.02$ threshold are reported explicitly in the SI.

At the individual marker level, four canonical stress response genes (DDIT3, ATF4, XBP1, HSPA5) were assessed using the same partial correlation framework.

2.11 Unbiased discordance GSEA (Approach B)

Approach A is the magnitude-conditioned Hallmark partial correlation above. Approach B asks whether unbiased Hallmark enrichment recovers apoptosis or p53 among terms associated with residual discordance. S_p is rank-residualized on magnitude; perturbations are split into residual quartiles; each discordant (Q4) perturbation is caliper-matched to a concordant (Q1) neighbor on magnitude. The matching gate requires $|SMD| \leq 0.25$ for magnitude, n_{cells} , depth, gene detection, and mitochondrial fraction, and at least 15 pairs. Cells per perturbation are equalized, profiles are pseudobulked, and Mann-Whitney tests are run across perturbations (not cells). Pre-ranked GSEA uses Hallmark in the main report and

KEGG/Reactome in the SI. Only Replogle and Norman are powered for this design. In the frozen run, magnitude balanced after matching, but cell-quality covariates failed the pre-specified gate in both datasets. Approach B was therefore dropped and Approach A carries the pathway result. This failure independently supports conditioning the primary partial models on cell-quality covariates; full SMDs are reported in **SI Appendix, Table S26**.

2.12 scGPT validation

To test whether the magnitude–coherence relationship is specific to the frozen linear embedding, we recomputed Sheshap coherence and magnitude in scGPT “Whole Human” pretrained embeddings (Cui et al. 2024) for four datasets (Norman, Adamson UPR, Dixit, Replogle). Raw counts (not log-normalized) were embedded using `embed_data()` with deterministic settings. Coherence and magnitude were computed identically to the frozen linear-embedding pipeline. The within-scGPT S_p –magnitude correlations are a redundancy check on Figure 2 and are not conflated with cross-representation concordance (**SI Appendix, Fig. S7**; embedding comparison against frozen S_p is **SI Appendix, Table S24**). Papalexi is excluded because scGPT ranks are not numerically reproducible at $n = 24$; the Adamson pilot is excluded ($n = 8$). Within-scGPT S_p –magnitude values are in **SI Appendix, Table S23**.

2.13 Diffusion maps and PHATE

To test whether S_p rankings are specific to the frozen linear embedding, we recomputed S_p and magnitude on the same frozen cells in scanpy diffusion maps (Haghverdi, Büttner, and Fabian J. Theis 2015; Haghverdi, Büttner, et al. 2016) (neighbors on 100 PCs, $k = 15$; trivial DC0 dropped so scoring uses the informative components) and in PHATE (Moon et al. 2019) ($k = 5$, internal PCA = 100, SGD-MDS, with SMACOF as the convergence control when SGD does not converge). Manuscript concordance is the Spearman correlation of each embedding’s S_p (and, separately, its magnitude) against frozen linear-embedding S_p (respectively M_p). Within-embedding S_p –magnitude correlations are reported separately and are not conflated with cross-representation concordance. Papalexi is excluded from scGPT concordance because S_p ranks are not numerically reproducible at $n = 24$. Per-dataset values and the PHATE convergence control are in **SI Appendix, Tables S24 and S25**.

2.14 Statistical analysis

All confidence intervals were computed via bootstrap resampling (10,000 iterations, seed 320, percentile method). The magnitude–coherence scatter plots (**Fig. 2** and **SI Appendix, Fig. S1**) display a LOESS smooth (bandwidth fraction = 0.4) in place of a linear regression line to capture the nonlinear attenuation at low magnitudes visible in the data; reported Spearman ρ values are point estimates with 95% bootstrap CIs computed as described above. Cross-dataset generalization is the pooled Spearman after within-dataset z-scoring (**Fig. 2g**). The SI reports a linear slope on those same z-scored axes (**SI Appendix, Table S5**). That companion is not the random-intercept model of raw S_p on globally z-scored magnitude, spread, and cell count. Partial correlations between coherence and stress markers were computed controlling for effect magnitude using rank-based partial Spearman throughout (not Spearman correlation of raw-scale residuals; **SI Appendix, Table S11**). All p -values are two-sided unless otherwise noted. Software pins, configuration version stamps, and digest checks are listed in **SI Appendix, Tables S2 and S4**.

Generative AI Usage

Large language models (specifically the GPT, Claude, Gemini, and Grok families) were used to assist with factoring the initial code for reproducibility, code debugging, software dependency management, and text editing and polishing. All hypotheses, experimental designs, analyses, and interpretations were independently formulated and verified by the author, and the author assume full responsibility for all content and claims in this work.

3 Results

3.1 Coherence tracks effect magnitude across modalities and effect-size definitions

Sheshap coherence tracks effect magnitude across all six datasets (**Fig. 2**; inventory in **SI Appendix, Table S1**). Spearman correlations run from $\rho = 0.843$ in Dixit (95% CI [0.754, 0.902]) to $\rho = 0.977$ in Replogle ([0.974, 0.979]); the two largest

screens are $\rho = 0.977$ (Replogle, $n = 1,832$) and $\rho = 0.950$ (Norman, $[0.930, 0.961]$, $n = 236$). The Adamson pilot ($n = 8$) is shown for completeness and is uninformative at that sample size. The same pattern holds for CRISPRa, CRISPRi and CRISPR-Cas9 knockout, in K562, BMDCs and THP-1, and in screens that differ by two orders of magnitude in size. That spread of settings argues against an idiosyncrasy of any one experiment: the coupling looks like a property of perturbation geometry in single-cell expression space.

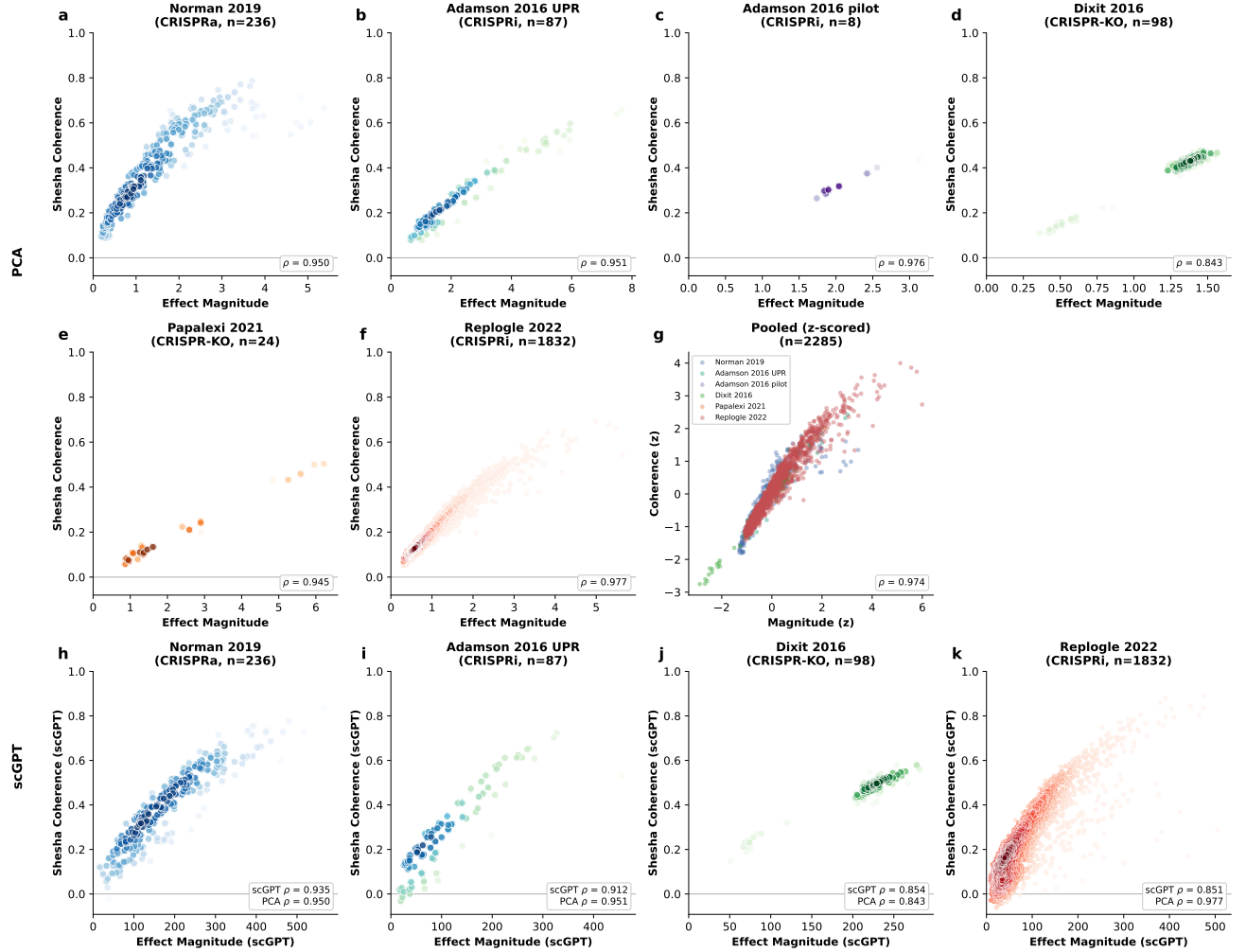


Figure 2: **Coherence tracks effect magnitude in PCA and in a foundation-model embedding.** Shesha_p (S_p) versus effect magnitude, with Spearman ρ annotated. Rows 1–2, frozen PCA: (a) Norman 2019 (CRISPRa), (b) Adamson 2016 UPR (CRISPRi), (c) Adamson 2016 pilot (CRISPRi; $n = 8$, uninformative), (d) Dixit 2016 (CRISPR-KO), (e) Papalexi 2021 (CRISPR-KO), (f) Replogle 2022 (CRISPRi), (g) pooled after within-dataset z-scoring. Row 3 recomputes both quantities in the scGPT embedding. The coupling survives that change of representation: ρ is 0.851–0.935 in scGPT versus 0.843–0.977 in PCA on the same four datasets (a different question from whether the two embeddings rank perturbations the same way; **SI Appendix, Fig. S7**). Papalexi and the Adamson pilot are omitted from row 3. Axes start at the origin; point color encodes per-panel perturbation density. Confidence intervals, sample sizes, and cell types are in Table 1.

Table 1: Directional coherence tracks effect magnitude across six CRISPR datasets. Spearman ρ between effect magnitude (Euclidean centroid displacement) and Sheshap coherence (S_p , mean cosine similarity of per-cell displacement vectors to the mean response direction) in PCA space. Bootstrap 95% confidence intervals (10,000 resamples, seed 320). Final row shows pooled (z-scored) results. These rank correlations do not depend on a fitted curve; LOESS fits are in **SI Appendix, Fig. S1**. ^a $n = 8$ is below the $n \geq 15$ threshold applied in the pathway analyses; reported here for completeness and excluded elsewhere. The upper confidence bound reaches 1.000 and the estimate should be treated as uninformative.

Dataset	Modality	Cell type	n	ρ	95% CI	p
Norman	CRISPRa	K562	236	0.950	[0.930, 0.961]	6.8×10^{-120}
Adamson (UPR)	CRISPRi	K562	87	0.951	[0.909, 0.972]	3.2×10^{-45}
Adamson (pilot) ^a	CRISPRi	K562	8	0.976	[0.795, 1.000]	3.3×10^{-5}
Dixit	CRISPR-KO	BMDC	98	0.843	[0.754, 0.902]	1.4×10^{-27}
Papalexi	CRISPR-KO	THP-1	24	0.945	[0.836, 0.984]	3.6×10^{-12}
Replogle	CRISPRi	K562	1,832	0.977	[0.974, 0.979]	$< 10^{-300}$
Pooled	—	—	2,285	0.974	[0.972, 0.976]	$< 10^{-300}$

Pooling all 2,285 perturbations after within-dataset z-scoring gave $\rho = 0.974$ ([0.972, 0.976]; **Fig. 2g**). A linear fit on those same z-scored axes has slope 0.956 (**SI Appendix, Table S5**). LOESS on the same points (**SI Appendix, Fig. S1**) shows that this rank correlation does not depend on a linear fit. Whitened (Mahalanobis) and k -NN matched local controls remain high across all six datasets (whitened $\rho = 0.963$ – 0.988 ; k -NN $\rho = 0.929$ – 0.988) and close the gap where Euclidean is weakest, most clearly in Dixit (0.843 versus 0.965 and 0.949; **SI Appendix, Fig. S2 and Table S6**). Embedding dimension from 10 to 100, leave-one-out deletion, analysis seed, and restriction to PS-called responder cells leave the same picture (**SI Appendix, Figs. S3, S4, S5 and Tables S7, S8, S9, S18**).

None of this is an artifact of how either quantity is computed. Recomputing both coherence and magnitude inside the scGPT foundation-model embedding still gives $\rho = 0.851$ to 0.935 on the four datasets with enough perturbations, against 0.843 to 0.977 in the frozen linear embedding (**Fig. 2h–k; SI Appendix, Table S23**). That within-scGPT redundancy is a different question from whether the two embeddings rank the same perturbations the same way (**SI Appendix, Fig. S7 and Table S24**). Defining effect size as energy distance—which additionally penalizes within-group dispersion—ranks perturbations almost identically to centroid displacement ($\rho = 0.950$ to 1.000) and accounts for slightly more coherence variance in every dataset (**Table 2; SI Appendix, Table S15**). The stricter test is an effect-size measure that shares no geometry with coherence: TRADE transcriptome-wide impact is estimated from gene-level differential expression with a power correction, and its correlation with coherence tracks its correlation with centroid displacement (median difference -0.01 ; Replogle 0.912 versus 0.919; **SI Appendix, Table S16**). Where transcriptome-wide impact diverges from displacement geometry, it diverges from coherence and effect magnitude alike.

Coherence and effect magnitude are tightly coupled, and that coupling is a property of the perturbation responses rather than of the representation or the effect-size definition. A large mean shift requires that individual cells move in a common direction, so perturbations that displace cells further from the control state necessarily do so more coherently. An isotropic Gaussian null already produces a near-perfect magnitude–coherence correlation through SNR alone, yet residual heterogeneity in the real data far exceeds that null (**SI Appendix, Fig. S6 and Table S10**). The rest of the paper asks what, if anything, coherence still carries once that coupling is accounted for.

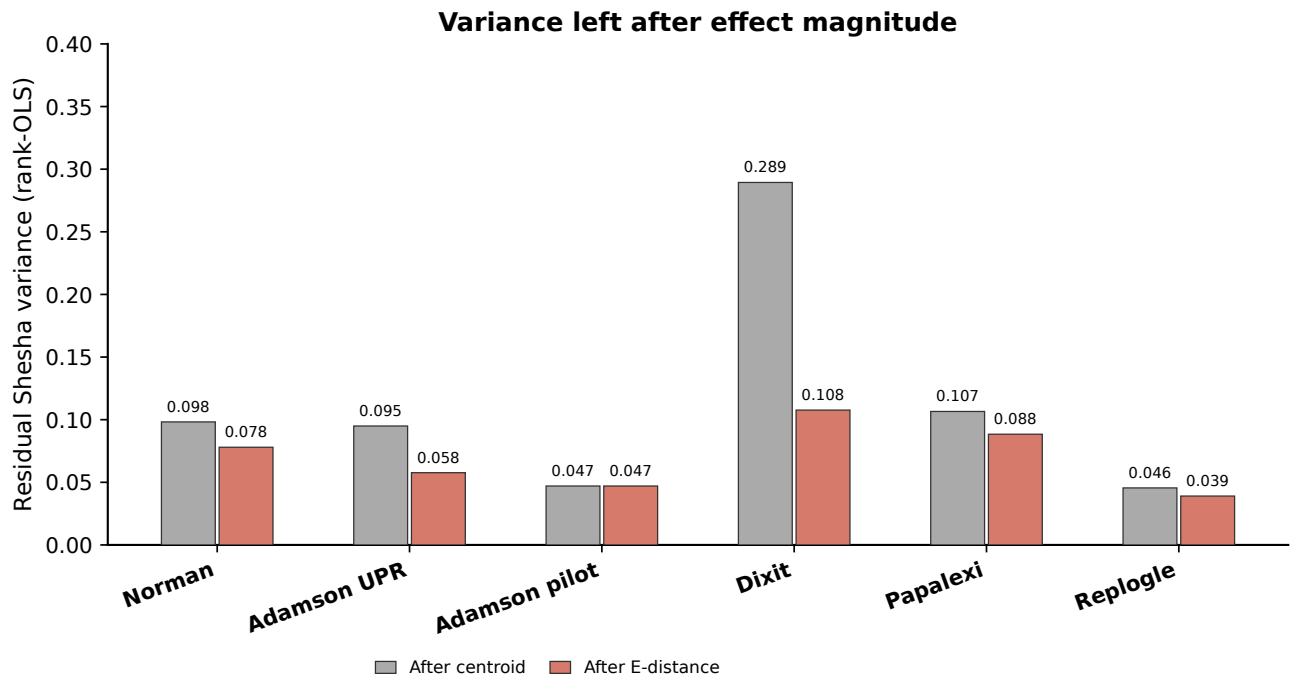


Figure 3: **Residual coherence variance after conditioning on effect size.** Fraction of Shesha variance unexplained by centroid displacement (grey) and by energy distance (red), rank-OLS. Energy distance explains more in every dataset, most sharply in Dixit (0.289 to 0.108), whose apparent excess residual under centroid displacement is largely within-group dispersion that energy distance captures. Conditional analyses therefore operate on 4–11% of coherence variance.

Table 2: Effect-size competitors on the frozen 50-PC embedding. ρ is Spearman; residual fractions are $1 - R^2$ from rank-OLS of $\text{rank}(S_p)$ on $\text{rank}(\text{size})$. Centroid displacement and energy distance are both computed from the same displacement vectors as Sheshap, so their high correlation with it is partly definitional. TRADE transcriptome-wide impact (TWI) is estimated from gene-level differential expression with a power correction and shares no geometry with coherence. The informative TRADE contrast is $\rho(S_p, \text{TWI})$ against $\rho(M_p, \text{TWI})$ rather than against $\rho(S_p, M_p)$: the two track closely (median difference -0.01 ; Replogle 0.912 vs 0.919), so where transcriptome-wide impact diverges from displacement geometry it diverges from coherence and effect magnitude alike. The Adamson pilot ($n = 8$) is uninformative.

Dataset	n	$\rho(S_p, M_p)$	$\rho(S_p, \text{Ed})$	frac cent.	frac Ed	$\rho(S_p, \text{TWI})$	$\rho(M_p, \text{TWI})$
Norman	236	0.950	0.960	0.098	0.078	0.688	0.812
Adamson UPR	87	0.951	0.971	0.095	0.058	0.954	0.968
Adamson pilot	8	0.976	0.976	0.047	0.047	0.857	0.833
Dixit	98	0.843	0.945	0.289	0.108	0.452	0.513
Papalexi	24	0.945	0.955	0.107	0.088	0.858	0.832
Replogle	1,832	0.977	0.980	0.046	0.039	0.912	0.919

3.2 Perturbation efficiency does not explain the coherence-magnitude relationship

Pooled CRISPR screens deliver perturbations of variable efficiency: within a population assigned the same guide, some cells respond and others do not. Unperturbed cells contribute short, arbitrarily oriented displacement vectors, so a guide pool with many non-responders will look both small and incoherent. That mechanism could produce the coupling above even if magnitude and coherence were unrelated among cells that actually respond.

We recomputed both quantities on Song et al. perturbation-response score callers only (Fig. 4a; SI Appendix, Table S18). Median responder fractions ran from 0.36 to 0.58, so 42–64 percent of assigned cells were called non-responding. Re-

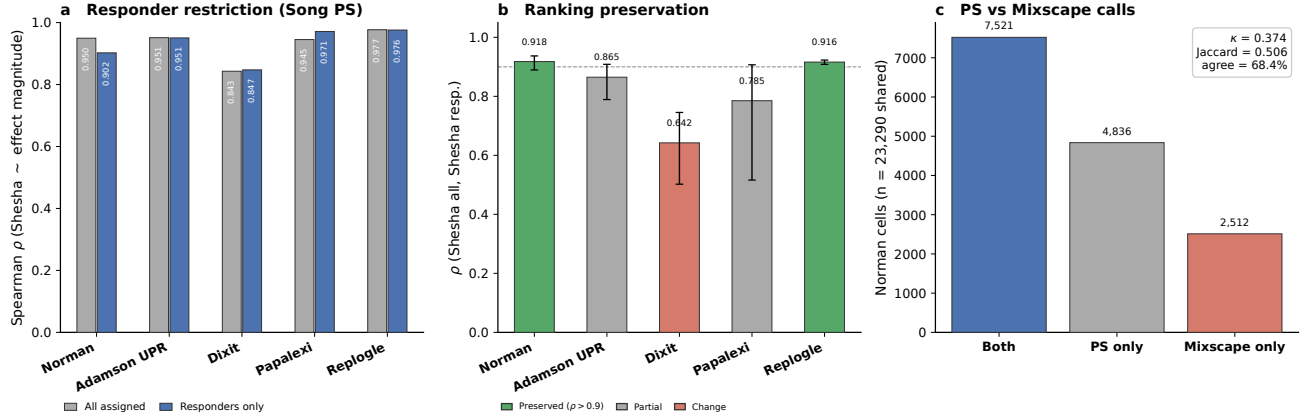


Figure 4: The coherence–magnitude relationship is not an artifact of variable perturbation efficiency. (a) Spearman ρ between Shesha_p and effect magnitude using all assigned cells (grey) and restricting to cells classified as responders by perturbation scoring (blue). Change ranges -0.048 to $+0.026$ across datasets, with no consistent direction. (b) Rank preservation between coherence computed on all cells and on responders only. Colors indicate the pre-specified verdict: preserved ($\rho \geq 0.9$), partial, or changed ($\rho < 0.7$). Dixit is the only dataset whose rankings change. (c) Per-cell responder calls in Norman disagree between the two standard classifiers ($\kappa = 0.374$, Jaccard = 0.506, 68.4% agreement on 23,290 shared cells), so responder-only reanalysis is conditional on the classifier and is reported as a sensitivity analysis rather than a correction. Error bars are bootstrap 95% CIs.

stricting to responders left the coherence–magnitude correlation in the same place in every dataset (**SI Appendix, Table S18**): Norman $0.950 \rightarrow 0.902$, Replogle $0.978 \rightarrow 0.976$, Adamson UPR $0.963 \rightarrow 0.951$, Dixit $0.841 \rightarrow 0.847$, Papalexix $0.945 \rightarrow 0.971$ (changes -0.048 to $+0.026$, no shared sign). These all-cell baselines are the efficiency-lane pair, not a substitute for the frozen Table 1 ρ . Rankings of perturbations were also largely preserved (**Fig. 4b**): Norman and Replogle sit above the pre-specified preservation threshold (rank $\rho = 0.918$ and 0.916); Dixit is the only dataset whose rankings move (rank $\rho = 0.642$). The coupling is not an artifact of unperturbed cells in assigned guide pools.

The same analysis also limits how far any responder-only reanalysis can be pushed. Mixscape on Norman, as a second caller, moved the correlation by $+0.030$ — opposite in sign to the perturbation–response score on that dataset, and on a 149/236 survivor subset rather than the frozen Norman baseline (**SI Appendix, Table S19**). Cell-by-cell comparison shows why: on the 23,290 Norman cells scored by both, the two callers agree on 68.4 percent of labels (Cohen’s $\kappa = 0.374$, Jaccard = 0.506; **Fig. 4c**; **SI Appendix, Table S20**). Mixscape is the more conservative caller (43.1 versus 53.1 percent responders) and is not nested inside the perturbation–response set: 2,512 cells are Mixscape-only. Which cells count as responders is method-defined, and the sign of the change in the coherence–magnitude correlation sits inside caller-to-caller variation.

Two limits follow. Responder-only reanalysis is conditional on a classifier and is a sensitivity check, not a correction of the primary estimates; every frozen value elsewhere in this paper uses all assigned cells. Both callers also select cells whose displacement aligns with an estimated perturbation signature, which is close to what coherence measures, so coherence rises on the restricted set by construction and we do not treat that rise as a result. The endpoints here are the coherence–magnitude correlation and the preservation of rankings, neither of which is fixed by the selection rule.

3.3 Concordance depends on the representation

Coherence is defined relative to a representation: the mean response direction, and each cell’s angle to it, are properties of the space in which displacement vectors are computed. We therefore asked how far perturbation rankings are preserved when that space changes, using three alternatives to the frozen linear embedding: diffusion maps, PHATE, and the scGPT foundation-model embedding (Cui et al. 2024). To separate a property of coherence from a property of perturbation geometry generally, we computed the same concordance for effect magnitude in each representation (**SI Appendix, Table S24**). Papalexix ($n = 24$) and the Adamson pilot ($n = 8$) are omitted from the scGPT arm: coherence ranks are not reproducible under GPU inference at those sample sizes (**SI Appendix, Fig. S4**).

Concordance depends on the representation for both quantities, and coherence is the more sensitive of the two. Under diffusion maps, coherence transfers well (Spearman $\rho = 0.641$ to 0.948 against frozen values) and matches or exceeds effect magnitude in four of six datasets. Under PHATE and scGPT, magnitude transfers better in every dataset tested. Across all sixteen dataset–representation comparisons, magnitude transfers at least as well as coherence in twelve.

The PHATE result is the most informative. Effect magnitude is largely preserved there ($\rho = 0.679$ to 0.891) while coherence is not, falling to 0.439 in Replogle and 0.506 in Papalexi. PHATE applies a potential-distance transform followed by metric embedding, a construction designed to linearize continuous trajectories. An embedding that preserves how far cells have moved while reorganizing which way they moved is precisely the case in which a displacement-direction statistic should fail to transfer, and that is what is observed. The pattern is not attributable to optimizer failure: MDS convergence does not predict concordance, with the three datasets whose embeddings converged spanning $\rho = 0.51$ to 0.68 and the three that did not spanning 0.44 to 0.88 (SI Appendix, Table S25). Under scGPT the two quantities degrade together, with gaps of 0.03 to 0.12 (Norman 0.514 versus 0.629 ; Replogle 0.750 versus 0.784 ; SI Appendix, Fig. S7), which points to the learned embedding not preserving displacement geometry for either statistic rather than to anything specific to coherence. That cross-representation question is separate from whether coherence and magnitude remain coupled *inside* scGPT, which they do ($\rho = 0.851$ to 0.935 ; Fig. 2h–k; SI Appendix, Table S23).

Two practical consequences follow. Coherence values should not be compared across representations, and in particular not across trajectory-linearizing embeddings, without establishing concordance first. And where a study reports coherence in a learned embedding, the effect-magnitude concordance should be reported alongside it, since without that comparator a moderate value cannot be attributed to the statistic rather than to the representation.

3.4 Residual coherence is associated with apoptosis and p53 signaling

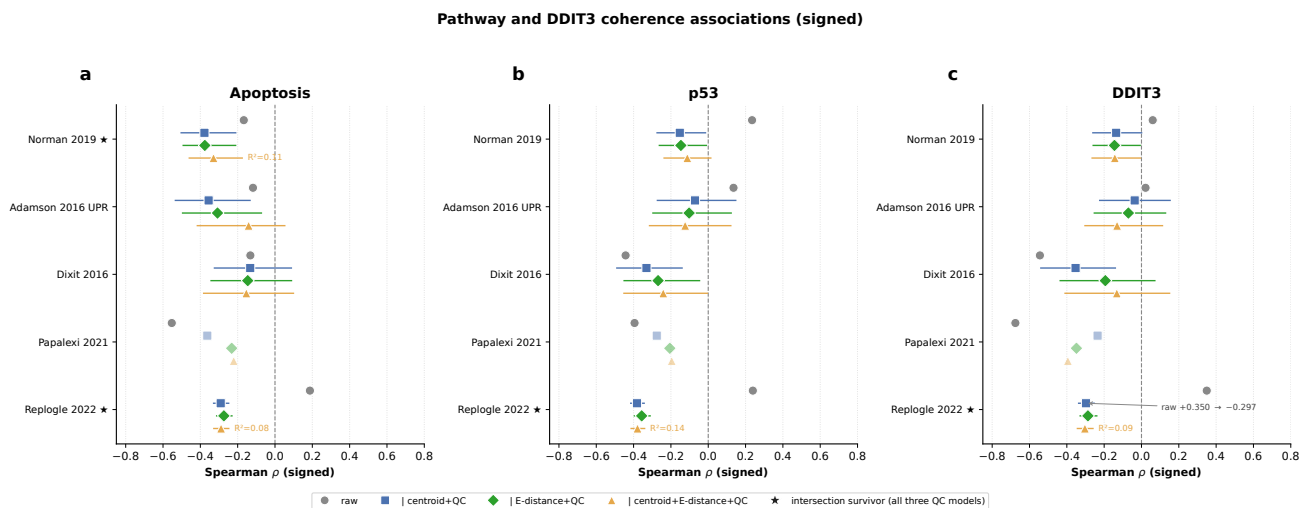


Figure 6: Associations surviving conditioning on effect size and cell quality. Signed Spearman ρ between Sheshap and (a) Hallmark Apoptosis, (b) Hallmark p53 Pathway, and (c) DDIT3 expression. Grey, unconditioned. Blue, green and gold, rank partials conditioning on centroid displacement, energy distance, or both, each with per-perturbation cell-quality covariates. Stars mark associations surviving all three models; partial R^2 is annotated on those rows under the joint model. Papalexi ($n = 24$) is point-estimate only. Most surviving associations reverse sign under conditioning, as expected when covariates explain 89–96% of coherence variance; Replogle DDIT3 is annotated as an example. Rows without stars include cases significant in some models but not all, such as Norman p53 and Dixit DDIT3. Values are in SI Appendix, Tables S13 and S12.

Coherence and effect magnitude are closely coupled, but the coupling is not complete: energy distance leaves four to eleven percent of coherence variance unexplained (SI Appendix, Table S15). We asked whether that residual carries biological signal, using composite scores for four MSigDB Hallmark gene sets (Apoptosis, p53 Pathway, Unfolded Protein Response, Reactive Oxygen Species) and four canonical stress markers.

Two considerations shape how this was tested. Residual coherence tracks cell quality: in a matched analysis, perturbations

in the lowest and highest residual-coherence quartiles could be balanced on effect magnitude but not on detected genes or UMI counts (**Fig. 7; SI Appendix, Table S26**). All partial correlations reported here therefore condition on per-perturbation mitochondrial fraction, detected genes, and total counts alongside effect size. Because these covariates are themselves partly downstream of apoptotic biology, conditioning on them removes real signal along with technical artifact, and the resulting estimates are lower bounds. Coherence is also coupled to effect size under more than one definition, so each association was tested against centroid displacement, energy distance, and both jointly. We report the intersection across all three models, never the union (**Fig. 6; SI Appendix, Tables S12 and S13**).

After conditioning, Apoptosis and p53 are negative in all five scoreable datasets. Apoptosis survives all three models in Norman ($\rho = -0.329$ under the joint model, partial $R^2 = 0.11$) and Replogle (-0.289 , $R^2 = 0.08$); p53 survives in Replogle (-0.379 , $R^2 = 0.14$). Among individual markers, DDIT3 is negative in all fifteen dataset-model cells and survives in Replogle (-0.303 , $R^2 = 0.09$). The two pathways are not redundant: their Hallmark gene sets share few members (Jaccard = 0.097) and their scores correlate weakly in every dataset contributing to the gate ($\rho = 0.145$ to 0.423 ; **SI Appendix, Table S14**).

Three features constrain what these associations support. Most reverse sign under conditioning: in Replogle, the unconditioned correlations are $+0.187$ for Apoptosis, $+0.239$ for p53 and $+0.350$ for DDIT3, all becoming negative once effect size is controlled (**Fig. 6**). This is expected when covariates explain 89 to 96 percent of coherence variance, but it means these are properties of the residual rather than of coherence as measured. Five of the six intersection survivors are in Replogle ($n = 1,832$); apoptosis replicates in Norman ($n = 236$) and no association survives the intersection in the three smaller datasets. This is therefore substantially a statement about what is detectable at that sample size, with the remaining datasets underpowered rather than negative. And the effects are small in absolute terms, operating on a residual of four to eleven percent of coherence variance.

The associations are also conditional on the family of effect-size measure. Conditioning instead on transcriptome-wide impact, estimated from differential expression rather than from displacement geometry, gives partial correlations of $+0.005$ for both Apoptosis ($[-0.040, 0.051]$) and p53 ($[-0.034, 0.045]$) in Replogle. The obvious explanation, that this measure is entangled with the pathway scores because both derive from expression, does not hold: its correlations with the two pathway scores (0.203 and 0.260) are lower than those of centroid displacement (0.254 and 0.327), and it is less correlated with coherence than centroid displacement is (0.912 versus 0.977 ; **SI Appendix, Tables S16 and S17**). It should be the more conservative covariate and it produces nothing. The associations therefore survive conditioning on two geometric effect-size measures and on cell quality, and do not survive conditioning on a differential-expression-derived measure of transcriptome-wide impact.

Among the remaining markers, ATF4 illustrates why individual genes are unreliable here. It survives in Replogle at $\rho = -0.106$ to -0.111 , marginal against the $|\rho| > 0.1$ threshold, while in Adamson UPR it survives with the opposite sign ($+0.340$ and $+0.315$ under two of three models). Both are CRISPRi screens in K562, so modality and cell type do not account for the reversal; screen composition is the more likely explanation, since Adamson UPR is a targeted 87-perturbation screen and Replogle is genome-scale. XBP1 survives in Replogle only, with positive point estimates in two other datasets. HSPA5 survives in no dataset under any QC-conditioned model (**SI Appendix, Table S12**). We do not treat any of them as a marker of residual coherence.

3.5 Raw associations are dominated by effect size

The preceding sections each conditioned on effect size before reporting an association. Collecting those comparisons alongside two others makes the pattern visible: unconditioned correlations with coherence are substantially or entirely accounted for by how large the perturbation's effect was (**Table 3**).

Two of the five vanish under conditioning. In Norman, combinatorial perturbations appear far more coherent than single-gene perturbations ($\rho = +0.424$; $n = 131$ versus 105), a difference that does not survive controlling for effect magnitude (-0.048 , $[-0.176, 0.091]$; adding cell-quality covariates and cell count gives -0.030 , $[-0.162, 0.113]$). The mechanism is transparent: combinatorial perturbations have substantially larger effects (magnitude standardized mean difference $+0.924$, with interquartile ranges overlapping by only 0.139), and coherence tracks magnitude at $\rho = 0.950$ in this dataset (**Table 1**), so the observed difference is what the effect-size gap alone predicts. Cell count does not account for it (standardized mean difference -0.37 ; single-gene perturbations are uniformly $n = 100$ cells). Norman also selected pairs expected to interact, which further limits generalization.

Table 3: Raw associations with Sheshap and the same associations after conditioning on effect size. Combinatorial and between-guide rows are rank-partial Spearman given centroid magnitude; Apoptosis, p53, and DDIT3 are Replogle centroid+QC (magnitude plus mitochondrial fraction and gene-count covariates), not the joint model in Fig. 6. In every case the unconditioned association is dominated by effect size: two vanish under conditioning and three reverse sign and persist on the residual. ^a131 combinatorial and 105 single-gene perturbations. Combinatorial perturbations have substantially larger effects (magnitude SMD = +0.924, IQR overlap 0.139), and coherence tracks magnitude at $\rho = 0.950$ in this dataset, so the raw difference is what the effect-size gap alone predicts. Cell count is not the driver (SMD = -0.37; single-gene perturbations are uniformly $n = 100$ cells). ^bThe multi-guide subset of the 1,832 scored perturbations (**SI Appendix, Table S21**).

Comparison	Dataset	n	raw ρ	ρ effect size	95% CI
Combinatorial vs single-gene ^a	Norman	131 / 105	+0.424	-0.048	[-0.176, 0.091]
Between-guide agreement ^b	Replogle	104	+0.332	+0.006	[-0.220, 0.225]
Apoptosis	Replogle	1,832	+0.187	-0.291	[-0.329, -0.248]
p53	Replogle	1,832	+0.239	-0.382	[-0.415, -0.343]
DDIT3	Replogle	1,832	+0.350	-0.297	[-0.337, -0.253]

In Replogle the agreement between mean response directions estimated from independent guides targeting the same gene correlates with coherence at $\rho = +0.332$, and at +0.006 ([-0.220, 0.225]) once effect magnitude is controlled ($n = 104$ multi-guide genes of 1,832 scored perturbations; **SI Appendix, Table S21**). The same partial is null in Papalexi (+0.183, [-0.317, 0.644]; $n = 23$). Coherence carries no information about whether independent reagents produce agreeing responses beyond what effect size already provides.

The remaining three reverse sign and persist. Apoptosis, p53 and DDIT3 all correlate positively with coherence in Replogle before conditioning (+0.187, +0.239, +0.350) and negatively afterwards under the centroid-plus-QC model used in this table (-0.291, -0.382, -0.297; **Fig. 6; SI Appendix, Tables S12 and S13**). Those three values are the centroid-plus-QC partials, not the joint-model intersection quoted in the pathway section (-0.289, -0.379, -0.303); the sign and the conclusion are the same. Reversal of this kind is expected when the covariate explains most of the variance in the outcome, but it means these associations are properties of the residual rather than relationships a reader would find by correlating coherence with pathway activity directly.

Directional coherence in Perturb-seq data is largely a restatement of effect magnitude. Associations reported without conditioning on effect size will mostly recover that coupling rather than anything about response geometry. This holds whether the comparison is between perturbation classes, between independent reagents, or with pathway activity, and it is the reason every association in this paper is reported both raw and conditioned.

3.6 Limits of the measurement

Several analyses were run to completion and did not yield usable results. We report them here because they bound what directional coherence can resolve, and because two of them constrain how the primary estimates should be read.

3.6.1 Unbiased enrichment could not be balanced on cell quality

To identify biology associated with residual coherence without pre-specifying gene sets, we compared the lowest and highest residual-coherence quartiles by differential expression and enrichment, matching perturbations on effect magnitude with a caliper and requiring covariate balance before proceeding. Matching balanced effect magnitude in both datasets (standardized mean difference +0.013 in Replogle, -0.024 in Norman) but the pre-specified criterion failed on cell quality: detected genes remained imbalanced in both (-0.46 and -1.34) and UMI counts in Norman (-1.25; **Fig. 7; SI Appendix, Table S26**). The enrichment analysis is therefore not reported. Residual coherence remains associated with cell quality after effect magnitude is matched away, which is why the pathway partials reported above condition on quality covariates rather than on effect size alone.

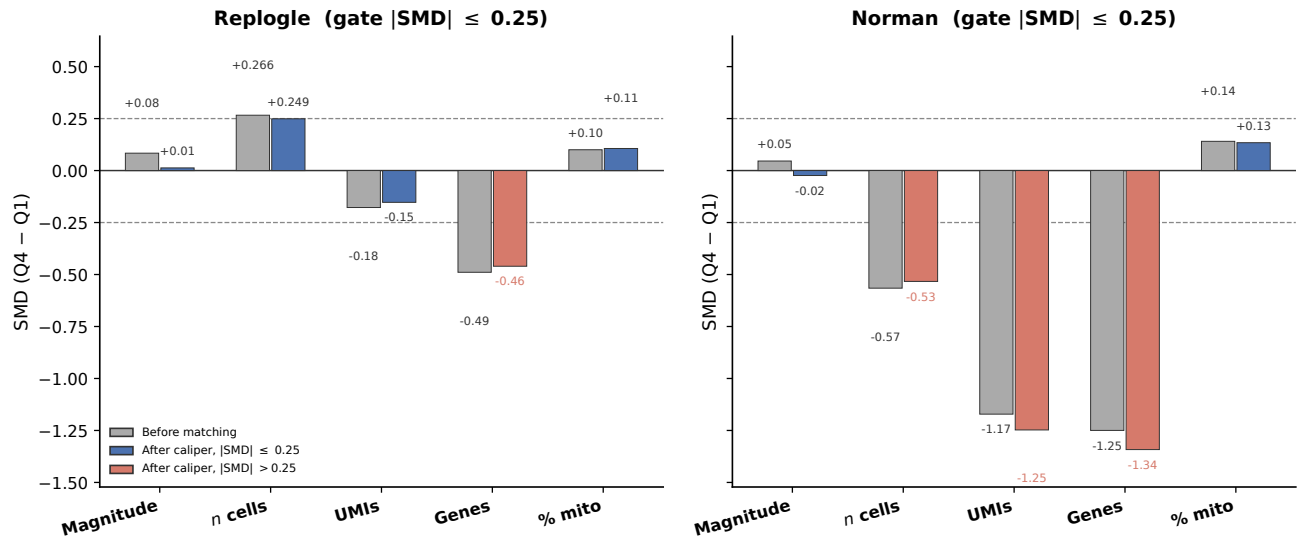


Figure 7: **Covariate balance in the discordance-quartile comparison, before and after caliper matching.** Standardized mean differences between the lowest and highest residual-coherence quartiles. Matching balances effect magnitude ($|SMD| \leq 0.25$, blue) but not cell quality: detected genes remain imbalanced in both datasets and UMI counts in Norman (red). The pre-specified balance criterion was not met, so the enrichment analysis is not reported. The residual association with cell quality is the reason primary partials condition on QC covariates. Full SMDs are in **SI Appendix, Table S26**.

3.6.2 A controlled simulation could not be calibrated to the observed range

We implemented a perturbation simulator to test whether coherence can distinguish responses that are heterogeneous because of variable perturbation efficiency from responses that are heterogeneous because cells engage different programs, a distinction the real data cannot separate. Across 120 simulated datasets and five seeds, the effect scale required to reproduce observed coherence proved dependent on which reference perturbation the pipeline selected, varying roughly fourfold at fixed settings when the reference changed. The generator also could not hold the magnitude of multi-program responses fixed while varying the number of programs, so the identifiability question it was built to answer was not testable as constructed. The arm is reported as attempted and not used (**SI Appendix, Table S27**).

3.6.3 The positive control does not clear conditioning

Perturbations targeting core unfolded protein response machinery in the Adamson UPR screen show lower coherence than the remainder (median 0.158 versus 0.268, $p = 0.013$), which is the expected direction for a perturbation class that should produce heterogeneous responses. Under the same survival criterion applied to the pathway analyses, the magnitude-conditioned partial is indeterminate ($\rho = -0.274$, $[-0.499, -0.008]$), with the nearer interval bound inside the knife-edge margin $\varepsilon = 0.02$. We were also unable to recover the gene set used in an earlier version of this analysis, so the previously reported value could not be reproduced. The control establishes a raw effect in the expected direction and does not establish one independent of effect size.

3.6.4 Interval estimation fails at small sample size

In Papalex (n = 24), the quality-covariate models are rank-deficient in bootstrap resamples and confidence intervals are not estimable (**SI Appendix, Tables S12 and S13**). Point estimates for that dataset are reported descriptively throughout and are excluded from every survival gate. The Adamson pilot (n = 8) is below the same $n < 30$ floor and is shown only for completeness.

3.6.5 Coherence rankings are not numerically reproducible at small n under GPU inference

Coherence is a rank statistic over closely spaced cosine values, so small floating-point differences reorder perturbations. Repeated scGPT inference on identical inputs changes concordance in the fifth decimal at $n = 1,832$ and in the first decimal at $n = 24$, where three runs on a fixed input gave 0.575, 0.633 and 0.630 (SI Appendix, Fig. S4 and Tables S4 and S8). Effect-magnitude concordance is exactly reproducible in both cases ($\rho = 0.753913$ on Papalexi). Deterministic algorithm settings reduced but did not eliminate the spread. Papalexi is therefore excluded from the scGPT concordance comparison, and coherence rank statistics computed on small perturbation sets under GPU inference should be treated as approximate.

4 Discussion

4.1 Coherence and effect magnitude

Directional coherence and effect magnitude are closely coupled in single-cell CRISPR screens. Across six Perturb-seq datasets spanning CRISPRa, CRISPRi and Cas9 knockout in three cell types, Spearman correlations run from 0.843 to 0.977 (Fig. 2; Table 1). The coupling is not an artifact of how either quantity is computed. It remains when both are recomputed in a transformer embedding (Cui et al. 2024), when effect size is an energy distance that also penalizes within-group dispersion (Peidli et al. 2024b; Székely and Rizzo 2013) (Fig. 3), and when effect size comes from gene-level differential expression rather than from displacement geometry (Nadig et al. 2025). Removing cells classified as non-responding does not remove it either (Song et al. 2025) (Fig. 4).

Unconditioned associations with coherence will therefore mostly recover effect size (Table 3). Combinatorial perturbations look more coherent than single-gene perturbations until magnitude is controlled; so does agreement between independent guides targeting the same gene. Three associations with stress-pathway activity reverse sign under the same step. A claim that one class of perturbation is more coherent than another, without that control, is usually a claim that one class has larger effects.

What is left after conditioning is small. Effect size accounts for 89 to 96 percent of coherence variance in the datasets used for the pathway analyses, a residual of four to eleven percent (Fig. 3). Inside that residual, coherence is negatively associated with apoptosis and p53 signalling and with DDIT3, an effector of the integrated stress response (Costa-Mattioli and Walter 2020) (Fig. 6). Those associations survive two geometric effect-size measures and per-perturbation cell quality, but they are concentrated in the largest screen, they are small, and they disappear when the covariate is transcriptome-wide impact. The residual is not empty. It is also not a reason to treat coherence as a substitute for effect size, or as a useful complement to it, in routine screen analysis.

4.2 Why the coupling is not a tautology

A close relationship between coherence and magnitude can look guaranteed. Coherence is the mean cosine of individual displacement vectors to their own mean—a mean resultant length in the sense of directional statistics (Mardia and Jupp 1999)—and the length of that mean is the effect magnitude. Vectors that agree in direction produce a longer mean than vectors that cancel. On that reading the correlation is arithmetic.

It is not forced. For a fixed set of displacement lengths the mean shift is longest when the vectors align and shortest when they cancel, so high coherence bounds magnitude from below. That bound runs only one way, and it says nothing about the joint distribution real perturbations occupy. Cells that all move a short distance in nearly the same direction give high coherence and low magnitude. Cells that move far in scattered directions give low coherence and high magnitude. Both are geometrically admissible, and both would be easy to interpret: a small, uniformly executed response, or a strong perturbation that different cells resolve differently through non-genetic heterogeneity or variable penetrance (Symmons and Raj 2016). The question is how often those configurations occur. In Perturb-seq they mostly do not; the data sit on the diagonal.

Two observations keep this empirical. The coupling is not equally tight everywhere: it is weakest in Dixit ($\rho = 0.843$) and strongest in Replogle (0.977), and the gap narrows when effect size is energy distance (0.945 and 0.980), so some of the between-dataset variation is within-group dispersion rather than the geometry of the mean shift (SI Appendix, Table S15). And the residual is not noise. If coherence were a deterministic function of magnitude, the leftover four to

eleven percent would be estimation error and would carry no structure. It carries associations with apoptosis and p53 signalling, in the same direction, in the two largest screens.

A quantity that could in principle vary independently of effect size largely does not, in this data. That is worth recording because the alternative looked plausible enough to motivate the measurement.

4.3 Relationship to existing measures

Nadig et al. 2025’s TRADE framework estimates transcriptome-wide impact from gene-level differential expression, with a correction for error in log-fold-change estimates. It shares no construction with coherence, so it is the cleanest check we have on whether the coupling is definitional. It is not. Coherence tracks transcriptome-wide impact about as closely as centroid displacement does (median difference -0.01 ; Replogle 0.912 versus 0.919; Adamson UPR 0.954 versus 0.968; **SI Appendix, Table S16**). Where transcriptome-wide impact diverges from displacement geometry, as in Dixit (0.452 and 0.513), it diverges from both quantities together. The one dataset where coherence loses more than magnitude is Norman (0.688 versus 0.812), the only CRISPRa screen. Breadth of effect across genes and agreement of direction across cells are different questions; in these data they are not independent of one another in the way that framing suggests.

Song et al. 2025’s perturbation-response score asks how strongly each cell responded, and we use it to classify responders (**Fig. 4**). A narrower point is worth keeping: per-cell distance from the control centroid is not a lightweight stand-in for that score. In Replogle the Euclidean and whitened centroid distances correlate with the algorithm-derived score at $\rho = 0.097$ and 0.149 (**SI Appendix, Table S28**). Any comparison of coherence to “response strength” that uses those proxies will recover the coherence–magnitude relationship, not a comparison to Song’s score, and we do not report one.

The two responder classifiers we compared do not identify the same cells. On the 23,290 Norman cells scored by both, they agree on 68.4 percent of labels ($\kappa = 0.374$; **SI Appendix, Table S20**). That disagreement has a counterpart in estimates of the unperturbed fraction in K562 Perturb-seq: Garbulowski et al. (Garbulowski et al. 2024) put it at roughly zero to 25 percent from simulated and experimental *E*-test distributions, while our perturbation-score responder fractions imply 42 to 64 percent non-responders (Papalexi et al. 2021). The two ranges are not reconcilable. What counts as a responding cell is method-defined, and any analysis restricted to responders inherits that.

Between-reagent consistency has been used as a quality filter outside transcriptomic screens, reflecting the well-documented variability of CRISPR guide efficiency (Doench et al. 2016). Seale et al. 2026 drop guides whose mutational-spectrum profiles correlate poorly with other guides on the same gene. The analogous question for coherence has a null answer once magnitude is in the model: in Replogle, gene-level coherence does not predict whether two guides agree on direction ($\rho = 0.006$, $[-0.220, 0.225]$). In Papalexi, guides targeting the same gene agree on coherence ($\text{ICC}(1) = 0.890$) and on effect magnitude to the same degree (0.889; **SI Appendix, Table S21**). Consistency in coherence is consistency in effect size.

4.4 The residual signal and its scale

After conditioning on effect size and cell quality, residual coherence is negatively associated with Hallmark Apoptosis and p53 signalling and with DDIT3 (**Fig. 6**; **SI Appendix, Tables S12 and S13**). Point estimates are negative in all five scoreable datasets for both pathways. Three associations survive every geometric covariate model: apoptosis in Norman and Replogle, p53 in Replogle. The two pathways are not one signal counted twice: they share 11 of 113 union genes, and their scores correlate only weakly in the datasets that enter the gate (**SI Appendix, Table S14**). DDIT3 is negative in all fifteen dataset–model cells and survives only in Replogle.

The direction matches a simple mechanistic sketch. Perturbations that leave cells in configurations no coordinated program resolves might engage apoptotic and p53-dependent responses (Tabas and Ron 2011; Vousden and Prives 2009), and coherence would then fall as those programs activate. The data are consistent with that sketch. They do not establish it.

Cas9-induced double-strand breaks activate a p53-mediated DNA-damage response (Haapaniemi et al. 2018; Ihry et al. 2018), so a p53 association could in principle be editing damage rather than perturbation response. Most of the surviving associations are not from nuclease screens. Both apoptosis survivors, and the Replogle p53 result, come from catalytically dead Cas9 (Norman CRISPRa; Replogle CRISPRi). Only Dixit p53 is from a Cas9 nuclease screen, and there the confound cannot be excluded.

Three limits follow. Effect size accounts for 89 to 96 percent of coherence variance, so the associations live on a residual of four to eleven percent, with partial R^2 of 0.08 to 0.14 inside that residual — under one percent of the variation in coherence, in absolute terms. Five of the six intersection survivors are in Replogle ($n = 1,832$), roughly eight times the next-largest screen; apoptosis replicates in Norman ($n = 236$), and nothing survives the intersection in the three smaller datasets. That is a statement about what is detectable at genome-scale n , not a claim that the smaller screens contradict it.

The associations also depend on how effect size is defined. They survive centroid displacement, energy distance, and both together, and they do not survive transcriptome-wide impact: in Replogle the partials for apoptosis and p53 are both +0.005, with intervals that include zero. That is not because TRADE merely restates the pathway scores. Its correlations with those scores (0.203 and 0.260) are lower than those of centroid displacement (0.254 and 0.327), and it is less correlated with coherence than centroid displacement is (0.912 versus 0.977; **SI Appendix, Tables S16 and S17**). It is a differently oriented covariate, not a more conservative one. Whether residual coherence carries stress-pathway signal therefore depends on which family of effect-size measure defines the residual.

Most of the survivors also reverse sign. In Replogle the unconditioned correlations are +0.187 for apoptosis, +0.239 for p53 and +0.350 for DDIT3, all negative once effect size is controlled. That is expected when the covariate explains most of the outcome and is itself correlated with the predictor. It also means these are properties of a residual, not relationships one would see by correlating coherence with pathway activity directly.

4.5 Implications for perturbation heterogeneity metrics

Single-cell CRISPR screens are increasingly summarized with statistics that go beyond mean effect size. Directional coherence is a natural one: it uses information that pseudobulk discards, it has a clear geometric definition, and it needs no extra experiment. Those properties are not enough. A statistic can be well defined, cheap to compute, and still mostly recover a quantity already in routine use.

Redundancy with effect size should be reported as a matter of course. We did not find a published perturbation-heterogeneity metric accompanied by its correlation with the corresponding effect magnitude; without that number a reader cannot tell whether a reported association belongs to the new statistic or to effect size measured another way. The same issue is familiar in expression variability, where cell-to-cell variation has to be separated from mean expression before it can be interpreted (Eling, Morgan, and Marioni 2019). The correlation is cheap.

One effect-size measure is not enough to claim independence. Centroid displacement and energy distance agree closely here ($\rho = 0.950$ to 1.000), and either alone would have supported the apoptosis and p53 associations; a differential-expression measure does not. A metric built from displacement geometry will tend to look independent of other geometric size measures. Testing against at least one measure constructed differently is what separates a shared-construction artifact from a finding.

Two of the analyses we ran—split-half agreement, and combinatorial versus single-gene coherence—give strong positive results that vanish under conditioning. Both are natural analyses, and both would have been easy to report without that step. Split-half agreement is circular by construction, a form of double dipping (Kriegeskorte et al. 2009): the halves come from the same cells and are compared to the same control reference, so a perturbation whose displacement vectors already align will produce agreeing half-samples regardless of reproducibility.

A more portable observation concerns efficiency filtering. Restricting to cells that responded to their assigned guide is common, and it depends on a classifier. On Norman the two standard callers agree on 68.4 percent of cells ($\kappa = 0.374$), and they move the coherence–magnitude correlation in opposite directions. Estimates of the unperturbed fraction in K562 Perturb-seq span roughly 0–25 percent in simulation-based work and 42–64 percent under the classifier used here. Results reported under a single caller, without that sensitivity, leave a reader unable to tell how much of the result is the classifier.

4.6 Context dependence

Several associations change sign between datasets, and where they do is informative.

Two reversals cross experimental context in the direction a context-dependent biology would predict. XBP1 is positively associated with residual coherence in Norman and negatively in Replogle (CRISPRa versus CRISPRi in the same cell line). mTORC1 and ROS reverse between Norman and Dixit, which differ in both modality and cell type. Gene activation and

gene interference engage different programs (Gilbert et al. 2013), and directional dependencies between the two have been shown in paired activation and knockout screens (Boettcher et al. 2018). We treat these as observations, not as isolated tests of modality: neither comparison holds the rest of the experimental design fixed.

ATF4 is a different case. It survives with a positive coefficient in Adamson UPR (+0.340 and +0.315 under two of three models) and a negative one in Replogle (−0.106 to −0.111). Both screens are CRISPRi in K562, so modality and cell type cannot explain the reversal. Adamson UPR is an 87-perturbation panel enriched for unfolded-protein-response machinery; Replogle is genome-scale. A marker’s relationship to residual coherence inside a pathway-focused panel need not match its relationship across a genome-wide set. That is a limit on generalizing from targeted screens, not an instance of biological context dependence, and we do not treat ATF4 as a marker of residual coherence (**SI Appendix, Table S12**).

Dixit is an outlier in more than one analysis: the weakest coherence–magnitude correlation (0.843), the largest residual after centroid displacement (0.289, falling to 0.108 under energy distance), the lowest concordance with diffusion-map and scGPT embeddings (**SI Appendix, Fig. S7 and Table S24**), and the only dataset whose perturbation rankings move under responder restriction (**Fig. 4**). It is also the only non-K562 dataset among the well-powered screens, and the earliest Perturb-seq experiment here. Its bone-marrow-derived dendritic cells were profiled under lipopolysaccharide stimulation, where paracrine signalling drives substantial cell-to-cell variation (Shalek et al. 2014). We cannot separate that biology from an early protocol, and we note the recurrence rather than attribute it.

Four of the six datasets are K562. The other two contribute 122 of 2,285 perturbations. Consistency across modalities is largely consistency within one cell line. Cancer cell lines differ in transcriptional state (Klijn et al. 2014) and diverge over passage and between laboratories (Ben-David et al. 2018), so even the K562 result should not be assumed to transfer.

4.7 Limitations

The dataset base is narrow in a specific way. Generality across cell types rests on Dixit ($n = 98$, BMDCs) and Papalexli ($n = 24$, THP-1). The surviving pathway associations are concentrated in Replogle; at $n = 1,832$ an association explaining under one percent of coherence variance is detectable, and at $n = 98$ it is not. The paper does not show that those associations generalize beyond genome-scale screens.

Conditioning on cell quality is conservative in a direction we cannot quantify. Mitochondrial fraction and detected-gene count are standard quality metrics but are partly downstream of cell state (Luecken and Fabian J Theis 2019), so regressing them out removes real signal with technical artifact, and the surviving estimates are lower bounds of unknown tightness. We condition on them because residual coherence remains associated with cell quality after magnitude is matched away (**Fig. 7**). The trade is between two biases.

In Papalexli ($n = 24$) the quality-covariate models are rank-deficient in bootstrap resamples, so that dataset contributes point estimates only and is excluded from every survival gate. Coherence rankings are also not numerically reproducible at that scale under GPU inference: three scGPT runs on identical inputs gave concordance 0.575, 0.633 and 0.630, while effect-magnitude concordance was bit-identical (**SI Appendix, Fig. S4**). Deterministic settings reduced the spread and did not remove it.

Coherence is a single scalar per perturbation. It does not capture subpopulation structure, bifurcating responses, or dose-dependent heterogeneity, and a bimodal mixture of strong responders and near-control cells can look like a unimodal response with the same mean angular spread. It is also defined relative to a representation, and rankings transfer imperfectly, least well under embeddings built to recover continuous trajectories (Saelens et al. 2019).

The associations are correlative. Showing that incoherent responses cause apoptotic or p53 signalling, rather than co-occur with it, would require manipulating coherence independently of magnitude—for example titrated knockdown with mismatched guides (Jost et al. 2020) or inducible depletion. We have not done that. Any such experiment would also need to specify in advance which effect-size measure it holds fixed.

4.8 Recommendations

The following are what we would tell a reader intending to compute directional coherence on their own data, or to evaluate a study that reports it.

4.8.1 Report the correlation with effect size, and condition on it before reporting anything else

In these datasets coherence and effect magnitude correlate at 0.843 to 0.977, so an unconditioned association between coherence and any other quantity is likely to be an association with effect size. This applies to comparisons between perturbation classes as much as to correlations with expression: two of the five comparisons we ran produced strong positive results that vanished entirely under conditioning.

4.8.2 Condition on cell quality as well

Residual coherence remains associated with mitochondrial fraction and detected-gene count after effect magnitude is balanced, so associations conditioned on effect size alone will partly reflect library quality. Conditioning on both is conservative, since these covariates are themselves partly downstream of cell state (Heumos, Schaar, et al. 2023), but the alternative is a confound rather than a clean estimate.

4.8.3 Test against an effect-size measure built differently

Centroid displacement and energy distance are both computed from the same displacement vectors as coherence, and they agree with each other closely here. A measure derived from differential expression rather than geometry gave a different answer for the pathway associations. Where a result depends on the effect-size family, that dependence should be reported rather than resolved by choosing the more favorable measure.

4.8.4 Do not use split-half agreement as a reproducibility endpoint

Both halves are drawn from the same cells and compared against the same control reference, so tightly aligned displacement vectors produce agreeing half-samples by construction. Independent reagents in independent cells are the minimum for a non-circular test, and even that shares the control reference.

4.8.5 Treat responder filtering as a sensitivity analysis, not a correction

Two standard classifiers agree on 68 percent of individual cells ($\kappa = 0.374$), and the direction of the effect on the coherence-magnitude correlation differs between them. Report results with all assigned cells as primary, with responder-restricted values alongside, and state which classifier was used.

4.8.6 Do not compare coherence values across representations without checking concordance

Perturbation rankings transfer imperfectly between embeddings, and least well under embeddings built to linearize continuous trajectories, where concordance fell to 0.44 while effect-magnitude concordance remained at 0.68. This is a specific instance of a general problem: low-dimensional embeddings distort the high-dimensional pairwise relationships they are derived from (Chari and Pachter 2023), and a directional statistic is more exposed to that distortion than a distance is. Where coherence is computed in a learned or nonlinear embedding, the corresponding effect-magnitude concordance should be reported alongside it, since without that comparator a moderate value cannot be attributed to the statistic rather than to the representation.

4.8.7 Interpret coherence rankings cautiously below roughly $n = 50$ perturbations

Confidence intervals for conditioned estimates were not estimable at $n = 24$, and coherence rankings at that scale were not numerically reproducible across repeated GPU inference on identical inputs. Effect-magnitude rankings were stable in the same comparisons.

4.8.8 Implementation

Sheshap is implemented in the open-source [shesha-geometry](#) package (Raju 2026c), which computes coherence and the effect-size measures used here from an AnnData object (Virshup et al. 2024) carrying perturbation labels and a control condition, together with the rank-based partial correlation and bootstrap routines these recommendations call for. The frozen analysis configuration, the per-dataset output tables, and the code reproducing every figure are available at [URL]; running the checks above on a new dataset requires no additional implementation.

Conflicts of interest

The authors declare that they have no competing interests.

Data availability

All analyses reported in this paper, including figure generation code, are available at <https://github.com/prashantcraju/shesha-crispr> and archived on Zenodo <https://doi.org/10.5281/zenodo.22116628> (Raju 2026b). All five datasets are publicly available through pertpy (Heumos, Ji, et al. 2025) and require no login or registration. The Shesha perturbation coherence metric is implemented in the open-source Python package shesha-geometry, freely available on PyPI (<https://pypi.org/project/shesha-geometry>; Raju 2026c) under an MIT license with accompanying tutorials.

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A Extended Methods

A.1 Datasets and preprocessing

Six single-cell CRISPR perturbation datasets were accessed via the `pertpy` Python package (version 1.0.4) (Heumos, Ji, et al. 2025). Table S1 summarizes the datasets. Modalities are CRISPRa (Norman), CRISPRi (Adamson UPR, Adamson pilot, Replogle), and CRISPR-KO (Dixit, Papalexi). The frozen Sp table comprises $n = 2,285$ perturbations; pathway and stress partials use five scoreable datasets ($n = 2,277$) after excluding the Adamson pilot ($n = 8$). Papalexi was additionally rebuilt from GEO (GSE153056) as an independent check that scores are not `pertpy-loader`-dependent. Each dataset was preprocessed independently to prevent batch effects. All manuscript-facing tables share one frozen preprocess and Sp-scoring path. The pipeline for each dataset consisted of:

1. Quality filtering: cells with fewer than 100 detected genes were removed.
2. Library-size normalization and `log1p` (Wolf, Angerer, and Fabian J. Theis 2018) were applied only when `.X` contained raw counts. Matrix scale was pinned per dataset from inspection of `.X`: Norman and Dixit are already log-transformed (normalize+`log1p` skipped); Adamson UPR/pilot, Papalexi, and Replogle are raw counts and were normalized to the median library size (`target_sum=None`) before `log1p`.
3. Frozen linear embedding: Adamson UPR/pilot and Papalexi used `scanpy.pp.highly_variable_genes(n_top_genes=2000, subset=True)` followed by `scanpy.pp.pca(n_comps=50)`. Norman, Dixit, and Replogle used variance/mean-ranked genes (all retained cells for Dixit; a seed-320 sample of 20,000 cells for Norman and Replogle) followed by `TruncatedSVD(n_components=50, random_state=320, n_iter=5)`.
4. Cell materialization: perturbations with fewer than 50 cells excluded; large groups downsampled in an order-invariant manner by hashing `seed|obs_name` (at most 100 cells per perturbation and 5,000 controls; seed 320).

All downstream coherence and magnitude computations were performed on the 50-dimensional frozen linear embedding.

A.2 Control group identification

Control group assignment used a multi-stage matching protocol to accommodate heterogeneous labeling conventions across datasets:

1. **Exact match** (case-insensitive): labels matching “control”, “ctrl”, “non-targeting”, “NT”, “unperturbed”.
2. **Delimiter-aware regex**: for short tokens (e.g., “NT” in “NTg1”), split on common delimiters and match components.
3. **Substring matching**: for longer keywords embedded in complex labels.

Dataset-specific handling:

- **Replogle 2022**: Labels containing “non-targeting” or beginning with “chr” were assigned to control. Labels containing “pos_control” were removed. Gene names were extracted by splitting on underscore and taking the first token.
- **Papalexi 2021**: The `gene_target` column was copied from MuData global metadata. Non-targeting guides (NTg1 through NTg7) were pooled into a single NT control group (2,386 cells). Gene-level Sp from the GEO rebuild matches the frozen gene-grain table (Spearman $\rho \approx 1$ on $n = 24$ shared perturbations).
- **Norman 2019**: The `perturbation_name` column was used directly. Cells labeled “control” served as the control group.
- **Adamson 2016 UPR**: Guide labels were aggregated to gene before scoring; Gal4 / mod non-targeting labels served as control.

A.3 Perturbation coherence and effect magnitude

For a perturbation p applied to n_p cells, let $\mathbf{x}_i \in \mathbb{R}^{50}$ denote the frozen linear-embedding coordinates of perturbed cell i and $\boldsymbol{\mu}_{\text{ctrl}}$ the centroid of all control cells in the same dataset. The shift vector for cell i is:

$$\mathbf{d}_i = \mathbf{x}_i - \boldsymbol{\mu}_{\text{ctrl}} \quad (2)$$

The mean perturbation direction is:

$$\bar{\mathbf{d}} = \frac{1}{n_p} \sum_{i=1}^{n_p} \mathbf{d}_i \quad (3)$$

Shesha perturbation coherence (S_p) is defined as:

$$S_p = \frac{1}{n_p} \sum_{i=1}^{n_p} \frac{\mathbf{d}_i \cdot \bar{\mathbf{d}}}{\|\mathbf{d}_i\| \|\bar{\mathbf{d}}\|} \quad (4)$$

Effect magnitude is:

$$M_p = \|\bar{\mathbf{d}}\| \quad (5)$$

The perturbation coherence metric adapts the principle of geometric self-consistency from the general Shesha framework (Raju 2026a,c,d). In that framework, Shesha-FS measures split-half RDM consistency across representations. Here, S_p measures perturbation coherence directly within a single representation, specializing the principle to perturbation biology.

A.4 E-distance competitor

E-distance (Peidli et al. 2024a) is scored on the same hash-selected cells and 50-PC embedding as S_p , using the full frozen control set (at most 5,000 cells). The implementation is the perpy Euclidean U -statistic with off-diagonal within-group means; for every dataset it is checked against an independent sklearn pairwise-distance reference. The signed estimate is retained and is never labelled magnitude. Residual S_p variance is $1 - R^2$ from rank-OLS of $\text{rank}(S_p)$ on $\text{rank}(\text{size})$. Method id: energy_distance_u_statistic_euclidean. Per-dataset correlations are in Table S15.

A.5 TRADE transcriptome-wide impact

TWI (Nadig et al. 2025) is $\text{Var}(\beta)$ from an ashr half-uniform mixture (Stephens 2016) fit to Welch $\log_2\text{FC}$ and SE on the frozen log-normalized transcriptome (same cells as S_p ; no HVG subset, no PCA). scanpy $\log_1\text{p}$ is converted from natural log to $\log_2$ by $/\ln 2$. Genes with non-finite SE, $|\log_2\text{FC}| > 10$, or matching the perturbed target are dropped. TWI values were computed via a Python implementation of the ashr half-uniform mixture variance (`mixcompdist='halfuniform'`, `prior='uniform'`), numerically consistent with R ashr (Stephens 2016). DE method id: welch_log2fc_on_frozen_log_transcriptome; fit id: trade_univariate_ashr_halfuniform. TRADE is a robustness column and is not folded into the three-model QC intersection. A diagnostic Spearman of TWI versus each Hallmark score (centroid magnitude as comparator) is required before interpreting TWI-conditioned pathway partials. The Adamson pilot ($n = 8$) is uninformative for pathway claims. Per-dataset correlations are in Table S16.

A.6 Minimum cell count filtering

Perturbations with fewer than 50 cells were excluded under the frozen filter. The Adamson pilot ($n = 8$) is retained in the six-dataset S_p table but excluded from pathway and stress partials.

A.7 Partial correlations

All partial correlations reported in the revision use rank-based partial Spearman correlation: variables are ranked before residualization (`pingouin.partial_corr`, `method='spearman'`, or an equivalent numpy `rank`→`residualize`→`Pearson` path). Legacy Spearman correlations of raw-scale residuals are not used.

A.8 Cell-quality covariates

Per-perturbation QC covariates are the mean of percent_mito, n_genes, and n_counts over the same frozen cells used for S_p . Mitochondrial percentage is taken from existing obs columns when present (fraction converted to percent if the maximum is ≤ 1.5) and otherwise computed from MT-genes. When $n < 40$, only mitochondrial percentage is used. Primary pathway and stress survivors are the intersection of |centroid+QC, |E-distance+QC, and the joint model.

A.9 Perturbation-efficiency sensitivity

To test whether S_p -magnitude redundancy is driven by unperturbed cells in assigned guide pools, S_p was recomputed after retaining only putative responder cells. The primary caller is Song et al.'s perturbation-response score (PS; within-perturbation rescaled $PS \geq 0.5$; computed via the Python reimplement of scMAGeCK described in Extended Methods); Mixscape is secondary (Norman only). Primary endpoints are (i) S_p -magnitude Spearman ρ before versus after filtering and (ii) rank correlation between S_p on all cells and S_p on responders. Absolute ΔS_p after filtering is logged but not interpreted as recovered coherence, because both callers select along the mean signature direction. The Adamson pilot was excluded by the same thin- n floor used elsewhere ($n < 30$); Papalexi ($n = 24$) is reported in **Table S18**. On Norman, per-cell PS-Mixscape agreement was modest (Jaccard = 0.506, Cohen's $\kappa = 0.374$), so "responder" is treated as method-defined and this arm is sensitivity-only (not a replacement for the frozen S_p table).

A.10 Association survival gate

Magnitude-conditioned and QC-conditioned associations were adopted when bootstrap CIs excluded 0, $|\rho| > 0.1$, and BH-FDR < 0.05 . Post hoc conservative rules: (i) demote rows whose nearer |CI| bound is strictly $< \varepsilon = 0.02$ to indeterminate; (ii) mito-only QC covariates when $n < 40$; (iii) point-estimate-only / exclude from the gate when $n < 30$. These post hoc rules and the adopted $\varepsilon = 0.02$ threshold are stated explicitly; no separate ε -sensitivity table is reported.

A.11 Unbiased discordance GSEA (Approach B)

The matched pseudobulk design and pre-specified balance criteria are described in Methods. In the frozen run, magnitude balanced after matching (SMD +0.013 in Replogle and -0.024 in Norman), but the gate failed on cell quality in both datasets (Replogle detected genes SMD -0.46; Norman UMIs and detected genes SMD < -1.2). Approach B was therefore dropped and Approach A carries the pathway result (Table S26). The independent persistence of cell-quality imbalance after magnitude matching supports including QC covariates in the primary partial models. Prior unmatched and cell-level GSEA outputs are retired.

A.12 Reproducibility

Frozen real-data tables share configuration version 2026-07-29.1 (random seed 320) and an Sp digest over (dataset, perturbation, stability, magnitude); the third field is the legacy schema name for Sheshap coherence, not a second metric. Downstream joins abort on stamp, row-count ($n = 2,285$), dataset-set, or digest mismatch. Frozen manuscript Sp is computed by local pipeline_core.calculate_sp. Auxiliary scripts that import shesha.bio use shesha-geometry==0.2.27 (commit dc0b4b1ea80e7b6ab76b61d4553dcd9757fd6aa1; <https://github.com/prashantcraju/shesha>). Simulation knobs are versioned separately (SYNTHETIC_CONFIG_VERSION=2026-07-25.9) and do not change real-data numbers; the synthetic arm is closed for manuscript use (SYNTH_MANUSCRIPT_USABLE=False). Responder-filter sensitivity outputs are stamped EFFICIENCY_FILTER_VERSION=2026-08-09.5 and analysis_role=sensitivity_only_not_a_freeze and do not replace the frozen Sp table. Retired real-data stamps (2026-07-25.1, 2026-07-17.5) must not be mixed with the current freeze. Calendar aliases are resolved before digest checks: 2026-08-04.1 maps to the current 2026-07-29.1 freeze with the same digest, whereas 2026-08-03.1 maps to the retired 2026-07-25.1 run and remains retired. See Table S2. E-distance scores use method energy_distance_u_statistic_euclidean on the frozen embedding. TRADE TWI uses welch_log2fc_on_frozen_log_transcriptome and trade_univariate_ashr_halfuniform (Python ashr implementation; numerically consistent with R ashr); downstream joins do not replace the frozen S_p table.

B Robustness Analyses

B.1 Distance metric robustness

All methods produce comparable or stronger correlations relative to the standard Euclidean metric (Table S6), confirming that the magnitude–coherence relationship is robust to the choice of distance metric. Whitened (Mahalanobis-scaled) and k -NN matched methods produce equal or higher correlations across all five primary datasets (Adamson, Dixit, Norman, Papalexi, Replogle), consistent with these methods reducing noise from batch effects and control group heterogeneity. Papalexi 2021 (CRISPR-KO, THP-1, $n = 24$) was analyzed with a minimum of 10 cells per perturbation to preserve sample size; all three metrics yield Spearman $\rho > 0.67$ with 95% bootstrap CIs excluding zero (Supplementary Figure S4). The Whitened method substantially improves the Dixit correlation ($\rho : 0.75 \rightarrow 0.97$), suggesting that residual covariance structure in the frozen linear-embedding space attenuates the Euclidean relationship in that dataset.

B.2 Linear-embedding dimensionality ablation

Coherence was recomputed at 10, 20, 30, 50, and 100 embedding dimensions, refitting each dataset’s frozen backend at every dimension (scanpy PCA for Adamson UPR/pilot and Papalexi; sparse TruncatedSVD for Norman, Dixit, and Replogle). The 50-dimensional rows reproduce the frozen main table rather than importing its values. For Replogle, Spearman ρ increased from 0.963 at 10 dimensions to 0.980 at 100 dimensions, with $\rho = 0.977$ at the frozen 50-dimensional setting. Results for all datasets are reported in Table S7.

B.3 Seed reproducibility

Cross-seed Spearman correlation of perturbation-level coherence rankings across 15 random seeds (320, 1991, 9, 7258, 7, 2222, 724, 3, 12, 108, 18, 11, 1754, 411, 103) was near-perfect: Norman mean $r = 0.99997$ (range [0.9999, 1.0000], 105 pairwise comparisons); Dixit mean $r = 0.99963$ (range [0.9993, 0.9999]). The magnitude–coherence Spearman ρ varied by less than 0.001 across seeds for both datasets (Norman: 0.959–0.960; Dixit: 0.844–0.844). All reported results use seed 320.

B.4 Leave-one-out influence

The maximum $|\Delta\rho|$ upon removing any single perturbation was 0.0019 for Norman (most influential: HES7; LOO ρ range [0.959, 0.961]) and 0.0106 for Dixit (most influential: INTERGENIC1144056+INTERGENIC1216445; LOO ρ range [0.840, 0.854]), indicating that no individual perturbation drives the overall correlation.

B.5 Theoretical null model

An isotropic Gaussian displacement of cells from a shared mean already produces a tight magnitude–coherence correlation, because a larger mean shift at fixed noise is a higher signal-to-noise ratio. To measure that null we simulated 2,000 perturbations in $d = 50$ dimensions ($\sigma \in \{0.5, 1.0, 2.0, 3.0\}$, 500 simulations per noise level). Coherence is almost perfectly predicted by SNR ($\rho = 0.999$; Fig. S6). After controlling for SNR the remaining partial correlation is $\rho_{\text{partial}} = 0.292$. Partial correlations in the real data, controlling for intrinsic spread and sample size, are larger in magnitude and opposite in sign across datasets: Norman $\rho_{\text{partial}} = -0.859$, Dixit $+0.627$ (Table S10). Residual heterogeneity in Perturb-seq therefore exceeds what SNR confounding produces on its own.

C Pooled linear companion

Cross-dataset generalization is the pooled Spearman after within-dataset z -scoring (Figure 2g; $\rho = 0.974$). Table S5 reports the linear companion on those same axes: both magnitude and Sheshap are z -scored within dataset, then $\text{sp}_z = \beta_0 + \beta_1 \text{mag}_z$ is fit with dataset fixed effects. The slope is 0.956 (SE 0.006). Dataset intercepts are ≈ 0 because the outcome was z -scored within dataset; that is not a random-effect variance estimate, and the table does not include spread or cell count. The Methods random-intercept specification (raw S_p on globally z -scored magnitude, spread, and $\log n_{\text{cells}}$) is a different model and is not the source of the numbers in Table S5.

D Extended Stress Marker Analysis

D.1 Partial correlations

Table S12 reports signed rank-partial Spearman correlations between perturbation coherence and four stress/UPR markers across the five scoreable datasets. The four models condition on magnitude, centroid magnitude plus QC, E-distance plus QC, or both size measures plus QC. Partial correlations use the rank-based procedure described in Extended Methods.

D.2 Modality analysis

After conditioning on centroid magnitude and QC, DDIT3 estimates were negative in all five scoreable datasets spanning CRISPRa, CRISPR-KO, and CRISPRi: Norman $\rho = -0.136$ [95% CI $-0.261, 0.001$], Adamson UPR $\rho = -0.036$ $[-0.224, 0.153]$, Dixit $\rho = -0.353$ $[-0.539, -0.140]$, Papalexi $\rho = -0.235$ (descriptive point estimate only at $n = 24$), and Replogle $\rho = -0.297$ $[-0.337, -0.253]$. Norman’s rounded centroid+QC interval extends to $+0.001$ and therefore crosses zero; its E-distance and joint upper bounds (-0.007 and -0.005) also lie within the pre-specified $\varepsilon = 0.02$ knife edge, yielding an indeterminate classification. Only Replogle passed the intersection criterion across all conditioned models. Thus the frozen analysis does not support modality as an explanation for DDIT3 heterogeneity. HSPA5 did not pass the intersection criterion in any dataset. Full results are provided in **Table S12**.

E scGPT Validation Protocol

E.1 Model and embedding

Cell embeddings were generated using the scGPT “Whole Human” pretrained checkpoint (scGPT_human), downloaded from the official repository (<https://github.com/bowang-lab/scGPT>). The checkpoint contains three files: `best_model.pt`, `vocab.json`, and `args.json`.

Input data consisted of raw counts (the counts layer was extracted from each AnnData object; data were not log-normalized prior to embedding). Embeddings were generated using `embed_data()` with `gene_col="index"`, `batch_size=64`, and `use_fast_transformer=False`.

E.2 Reproducibility settings

All scGPT computations used deterministic mode:

- Python random seed: 320
- NumPy random seed: 320
- PyTorch manual seed: 320 (CPU and CUDA)
- `torch.backends.cudnn.deterministic = True`
- `torch.backends.cudnn.benchmark = False`

E.3 Coherence and magnitude computation

Coherence and magnitude were computed from scGPT embeddings using `shesha.bio.compute_stability()` and `shesha.bio.compute_magnitude()` with `perturbation_key='perturbation_name'`, `control_label='control'`, and `metric='cosine'` (coherence) or `metric='euclidean'` (magnitude). Bootstrap confidence intervals: 10,000 resamples, seed 320, percentile method. The first function retains its historical API name; its returned mean-cosine quantity is Sheshap coherence throughout this manuscript.

E.4 Datasets

scGPT (Cui et al. 2024) analysis was performed on Norman 2019, Adamson 2016 UPR, Dixit 2016, and Replogle 2022. Control-labeled Adamson rows were excluded, leaving $n = 87$ scored perturbations. The Adamson pilot was omitted because $n = 8$; Papalexi was omitted because scGPT ranks were not numerically reproducible at $n = 24$.

E.5 Results

Within-scGPT magnitude–coherence correlations were 0.935 (Norman), 0.912 (Adamson UPR), 0.854 (Dixit), and 0.851 (Replogle). Cross-representation concordance against the frozen linear embedding was, respectively, 0.514, 0.828, 0.472, and 0.750 for coherence and 0.629, 0.873, 0.549, and 0.784 for magnitude. Because both score types change across representations, we interpret this as embedding sensitivity and do not assign it a coherence-specific biological mechanism. See [Tables S23 and S24](#).

E.6 Diffusion maps and PHATE

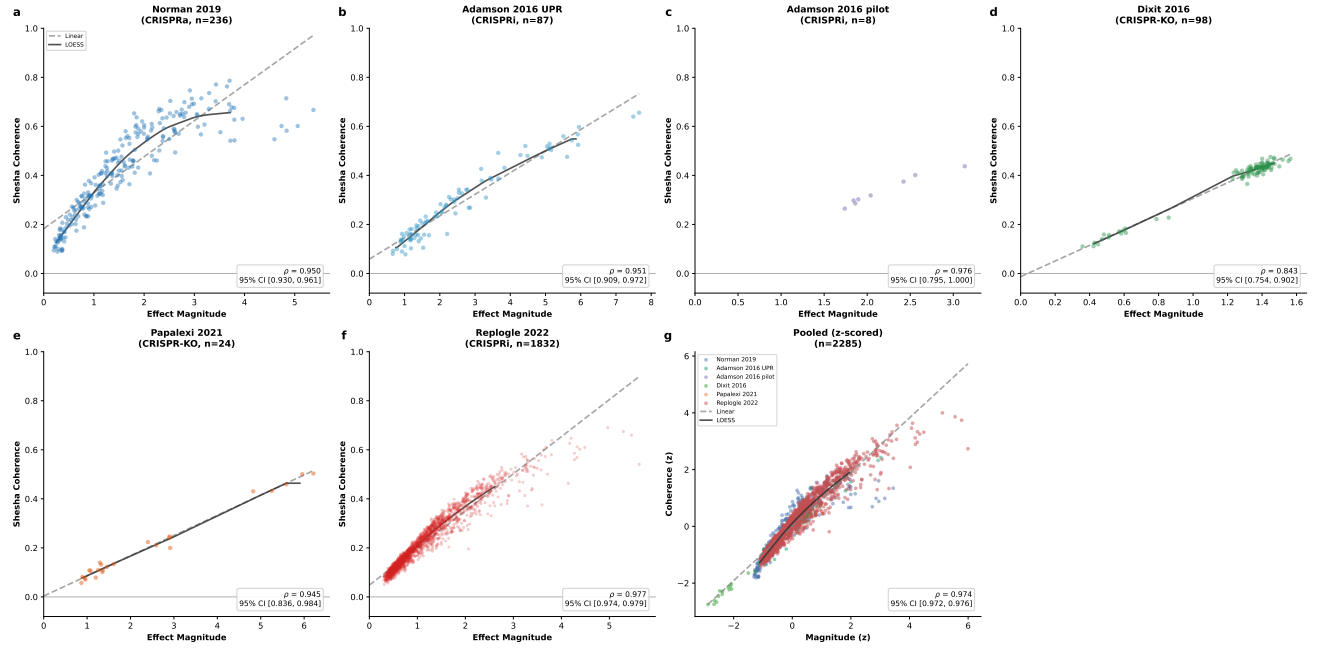
Diffusion maps used scanpy neighbors on 100 PCs ($k = 15$) with the trivial DC0 component dropped (Haghverdi, Büttner, et al. 2016). PHATE used $k = 5$, internal PCA = 100, and SGD-MDS (Moon et al. 2019); SMACOF is the convergence control when SGD does not converge. Manuscript concordance is each embedding’s S_p (and separately its magnitude) versus frozen linear-embedding S_p (respectively M_p), not versus a re-derived linear-embedding score. Within-embedding S_p –magnitude correlations are a separate column. Papalex is excluded from scGPT concordance because ranks are not numerically reproducible at $n = 24$.

F Combinatorial Analysis

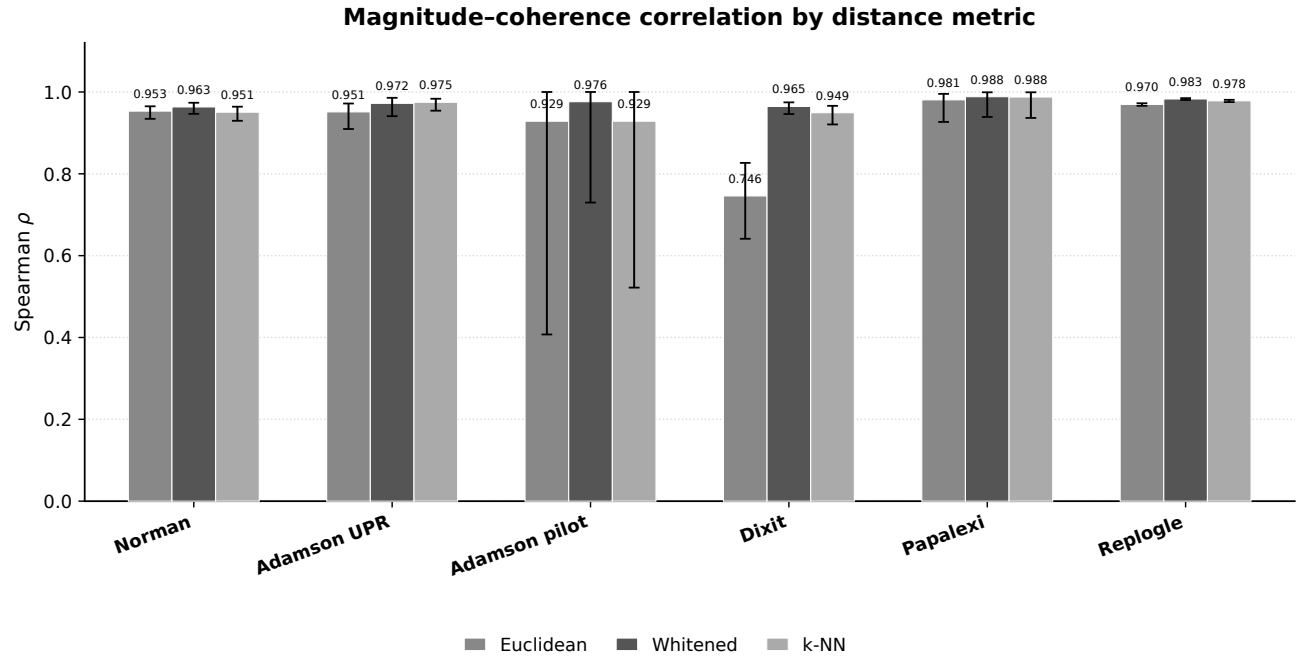
The Norman 2019 dataset contains $n = 105$ single-gene and $n = 131$ combinatorial perturbations. Combinatorial status was associated with higher raw coherence (mean $S_p = 0.490$ vs 0.334; Mann–Whitney $p = 8.4 \times 10^{-11}$; Spearman $\rho = 0.424$), but combinatorial perturbations also had larger effects (magnitude SMD = 0.924; IQR overlap = 0.139). After conditioning on magnitude, the association was null ($\rho = -0.048$, 95% CI $-0.176, 0.091$); adding QC and cell count gave $\rho = -0.030$ $[-0.162, 0.113]$. Thus the raw combinatorial association does not provide evidence for greater coherence beyond effect size. Norman selected gene pairs expected to interact, which further limits generalization. Within the combinatorial set, erythroid-containing pairs also had higher raw S_p (median 0.568 vs 0.465; one-sided Mann–Whitney $p = 0.0035$), but this contrast was null after magnitude residualization ($p = 0.518$).

G Perturbation-Efficiency Sensitivity

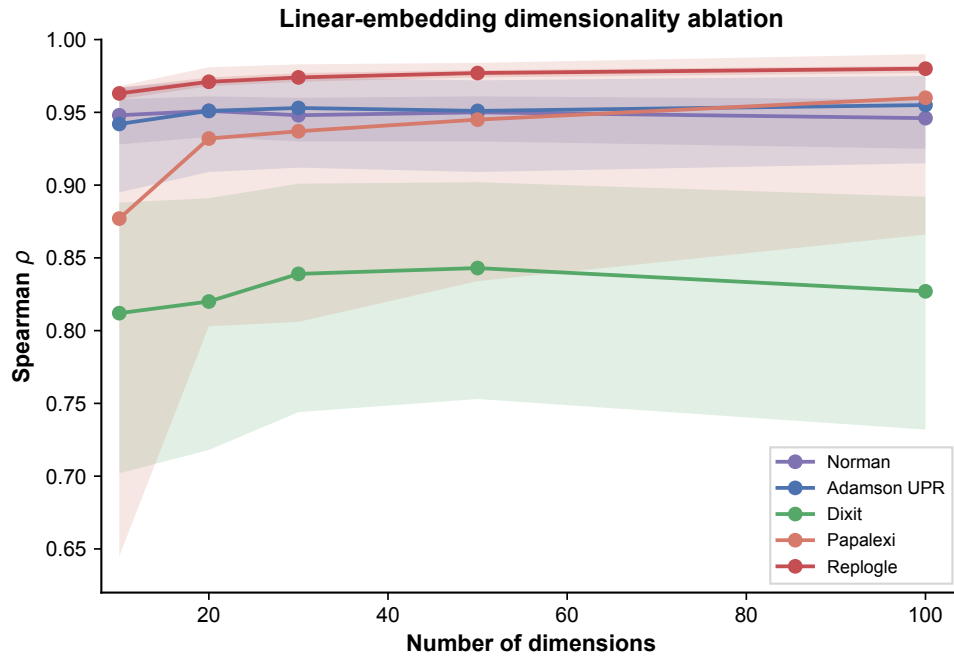
Primary arm (Song PS) on Norman, Replogle, Dixit, and Adamson UPR. On Norman and Replogle, S_p –magnitude correlations were essentially unchanged after responder filtering ($0.950 \rightarrow 0.902$; $0.978 \rightarrow 0.976$) and S_p rankings were preserved ($\rho = 0.918$ and 0.916). Adamson UPR was partial ($0.963 \rightarrow 0.951$; rank $\rho = 0.865$); Dixit changed more ($0.841 \rightarrow 0.847$; rank $\rho = 0.642$) and is the coherent outlier across efficiency filtering, residual S_p variance, and embedding concordance. Mixscape on Norman is secondary only: it rescored 149/236 perturbations (87 with no responders), so its “all-cell” S_p –mag baseline is a survivor subset rather than the frozen Norman value, and the S_p –mag Δ can flip sign relative to PS. Across callers, Δ in S_p –mag after filtering is within method-to-method noise; the relationship is not attributed to unperturbed cells in assigned guide pools.



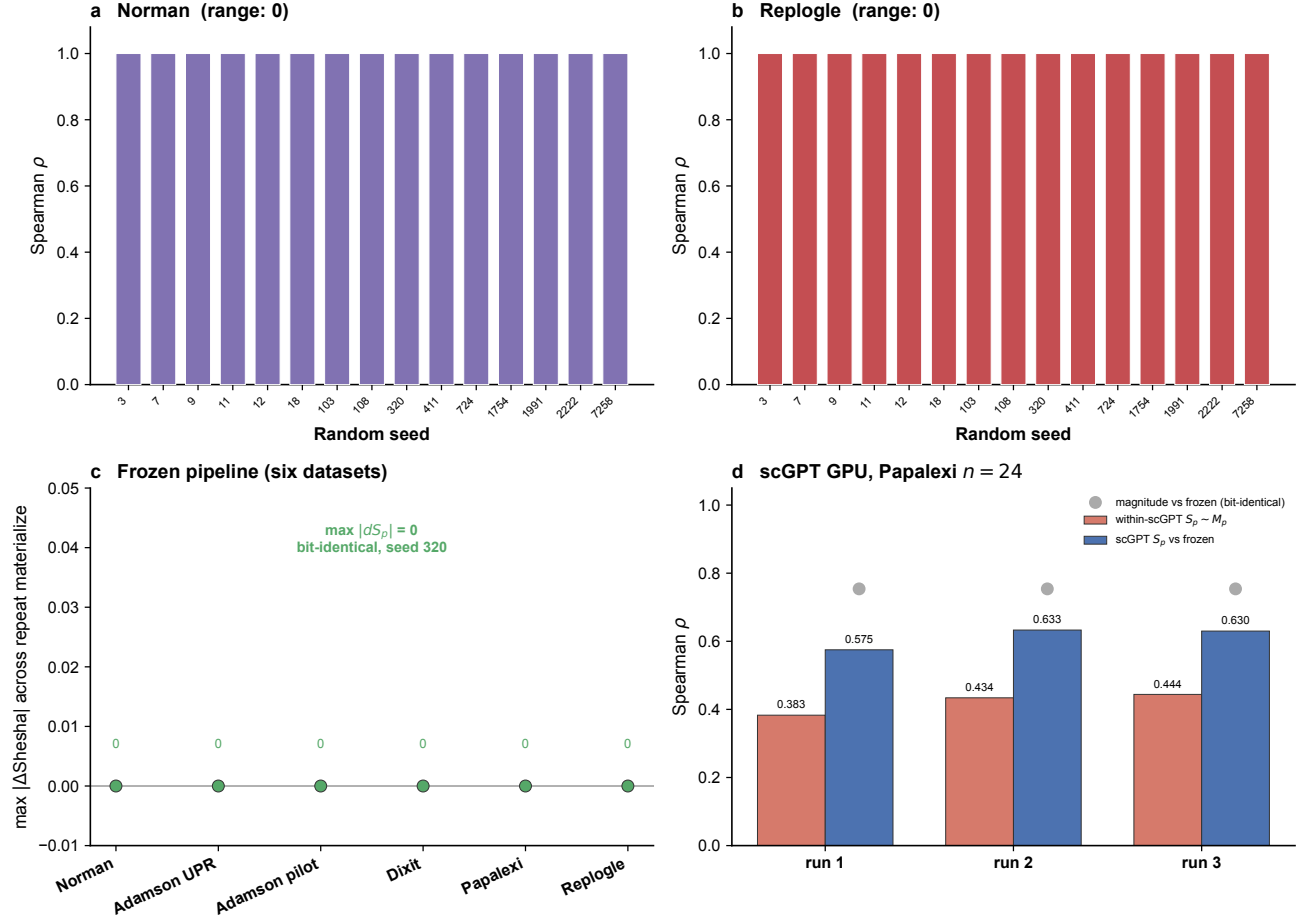
Supplementary Figure S1: **Coherence increases monotonically with effect magnitude in every dataset, with deceleration at high magnitude.** Shesha_P against effect magnitude for (a–f) each dataset and (g) pooled after within-dataset z-scoring. Dashed grey, ordinary least squares; solid black, LOESS. Deceleration is clearest in the well-powered datasets (f, g), consistent with coherence being a bounded cosine statistic, and the two fits are close in the smaller datasets. Reported ρ are Spearman rank correlations and are therefore independent of fit form; intervals are bootstrap 95% CIs. No fit is shown for the Adamson pilot ($n = 8$). LOESS is drawn only over the range where local density supports estimation.



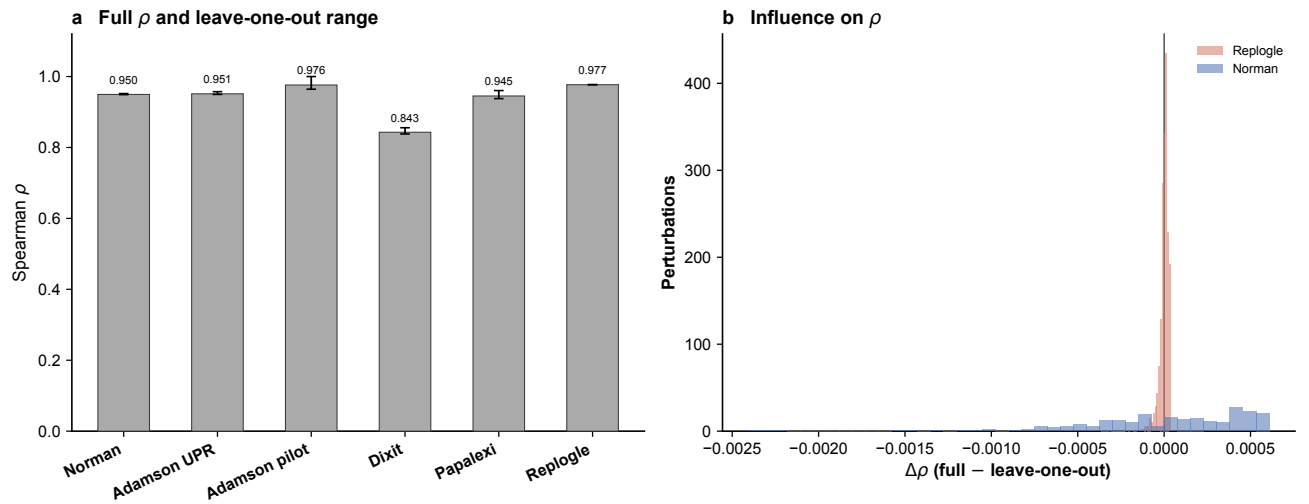
Supplementary Figure S2: **Magnitude-coherence is not an artifact of Euclidean PCA distance.** Spearman ρ between Shesha coherence and effect magnitude under three control-referenced metrics, six datasets: Euclidean centroid, whitened (Mahalanobis / $\Sigma^{-1/2}$), and k -NN matched local controls ($k = 50$). Bars are 95% bootstrap CIs.



Supplementary Figure S3: **Linear-embedding dimensionality ablation.** Magnitude–Shesha Spearman ρ as a function of embedding dimension (10, 20, 30, 50, 100) under 2026-07-29. 1. Shaded regions are 95% bootstrap CIs (10,000 iterations, seed 320). Each series refits the recorded frozen backend (scanpy PCA or sparse TruncatedSVD) on the freeze cell set; the 50-dimensional point reproduces Table 1. Norman ($n = 236$) is stable ($\rho = 0.946$ – 0.951) with overlapping CIs. Adamson UPR ($n = 87$) is likewise stable ($\rho = 0.942$ – 0.955). Dixit ($n = 98$) is lower throughout ($\rho = 0.812$ – 0.843) and does not increase monotonically. Papalexi ($n = 24$) rises from $\rho = 0.877$ to 0.960 with the widest intervals. Replogle ($n = 1,832$) remains high ($\rho = 0.963$ – 0.980). The choice of 50 dimensions does not drive the magnitude–Shesha relationship.

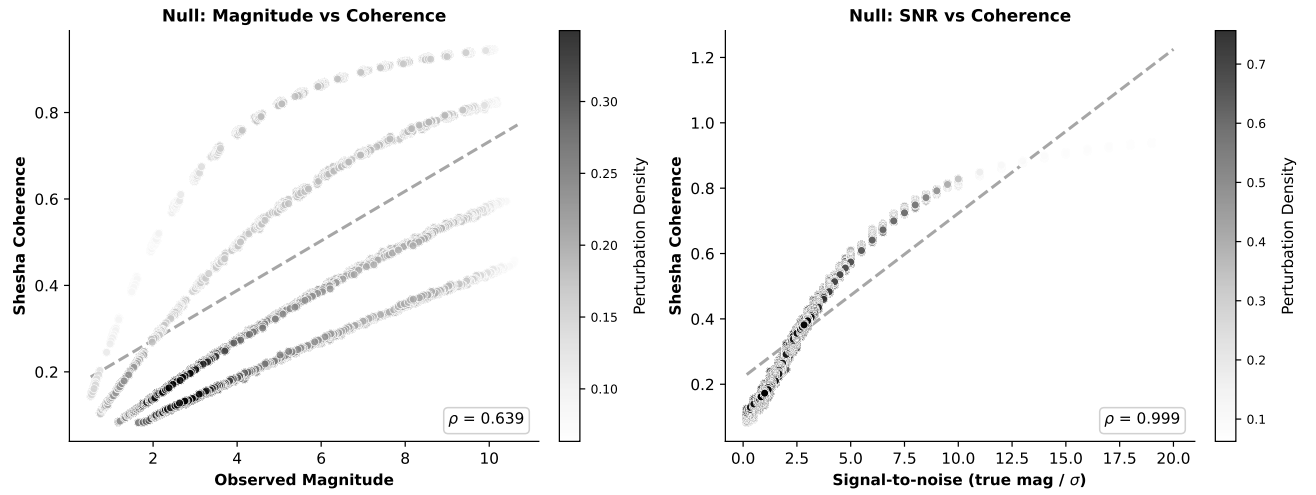


Supplementary Figure S4: **Seed and freeze reproducibility; scGPT ranks are not stable at small n .** (a,b) Magnitude-coherence Spearman $\rho = 1.000$ at 15 random seeds on Norman and Replogle (3, 7, 9, 11, 12, 18, 103, 108, 320, 411, 724, 1754, 1991, 2222, 7258); range = 0. (c) Repeated materialize + Shesha under the freeze (seed 320) yields $\max |dS_p| = 0$ on all six datasets. (d) Three locked-h5mu GPU scGPT embeds of Papalexi ($n = 24$): within-scGPT $S_p \sim M_p$ spans 0.383–0.444; concordance versus frozen S_p spans 0.575–0.633; magnitude concordance is bit-identical at 0.7539130434782607.



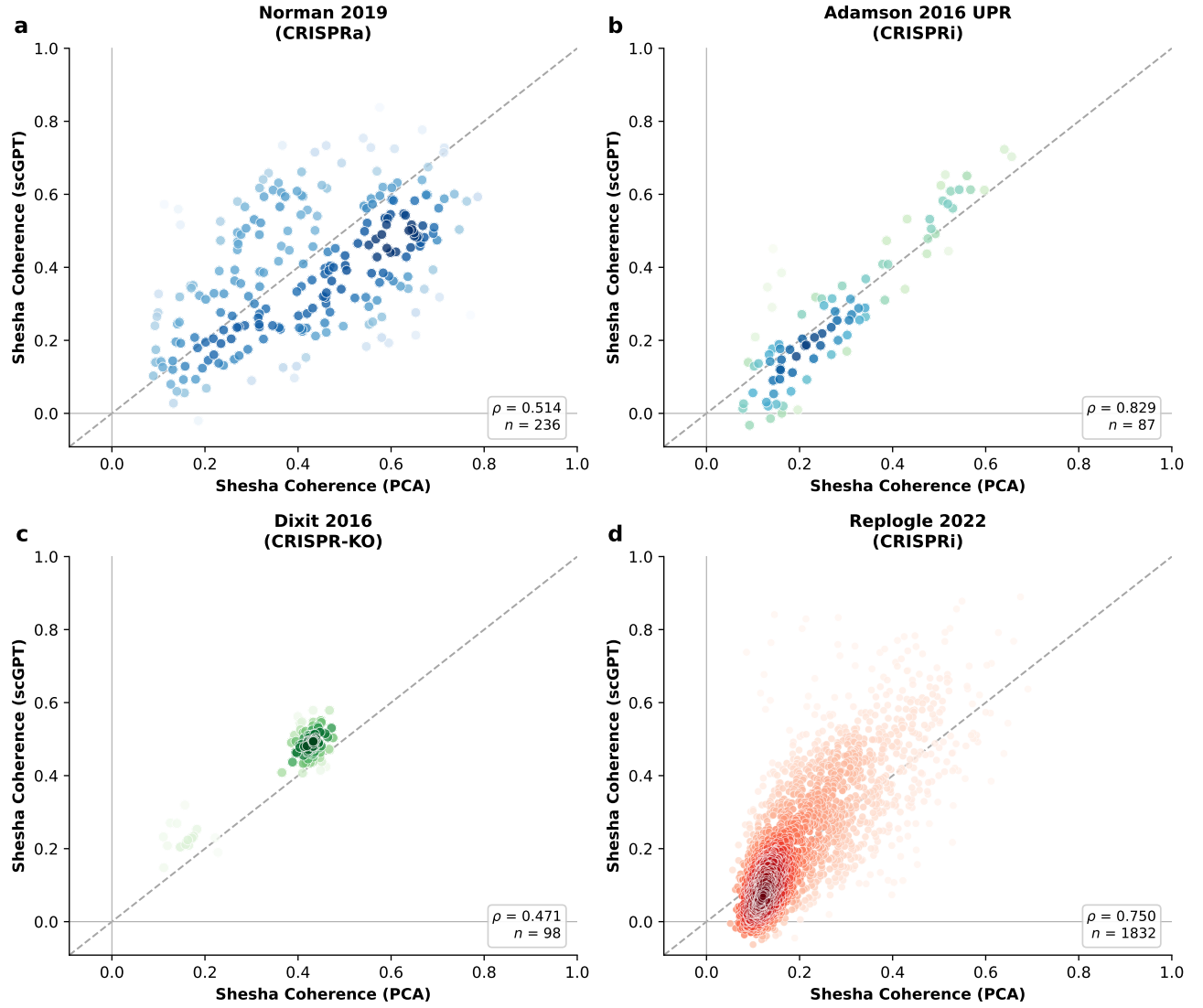
Supplementary Figure S5: **Leave-one-out influence on the magnitude-coherence Spearman ρ .** (a) Full ρ (bars) and the range of ρ after dropping each perturbation in turn (error bars), computed from frozen Shesha on all six datasets. Full ρ : Norman 0.950 ($n = 236$), Adamson UPR 0.951 ($n = 87$), Adamson pilot 0.976 ($n = 8$), Dixit 0.843 ($n = 98$), Papalexi 0.945 ($n = 24$), Replogle 0.977 ($n = 1832$). (b) Per-perturbation $\Delta\rho$ (full – leave-one-out) for Norman and Replogle. Maximum $|\Delta\rho|$ is 0.0024 on Norman and 0.0004 on Replogle; the largest swing among the six datasets is 0.024 on Adamson pilot ($n = 8$). No single perturbation drives the correlation.

Theoretical null model (isotropic Gaussian shift)

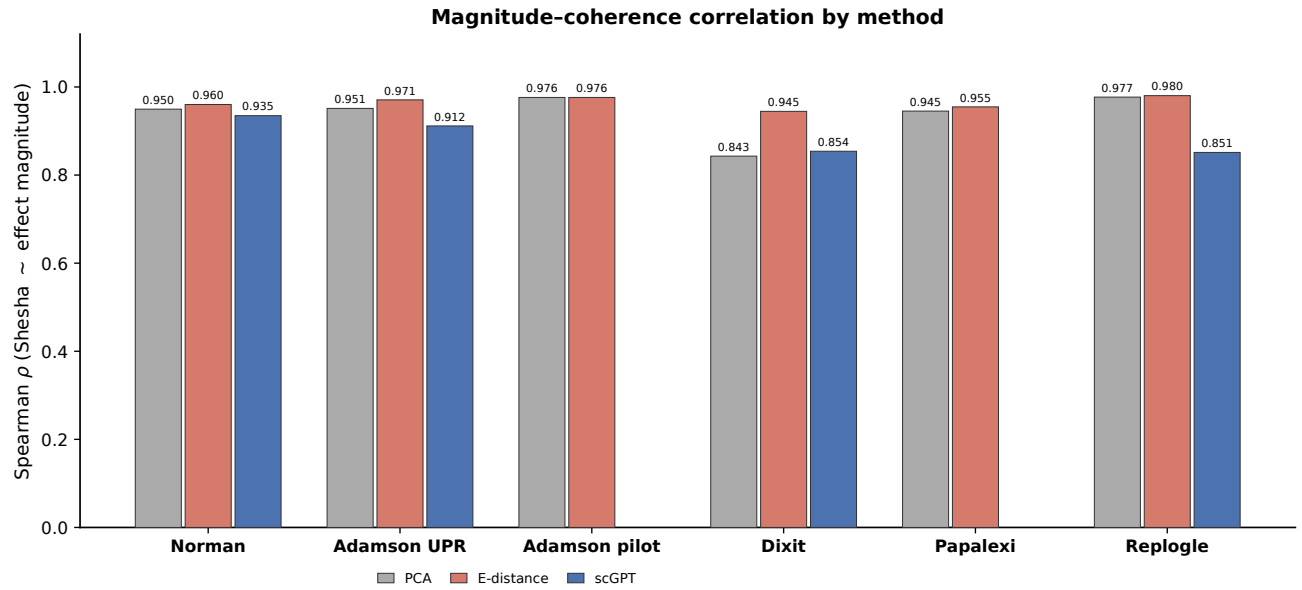


Supplementary Figure S6: **Theoretical null model under isotropic Gaussian perturbations.** Left: magnitude versus Shesha coherence. Right: the same simulations versus SNR. 2,000 simulated perturbations ($d = 50$ dimensions, $\sigma \in \{0.5, 1.0, 2.0, 3.0\}$, 500 simulations per noise level). Under the null model, coherence is almost perfectly predicted by SNR ($\rho = 0.999$), with a partial correlation of $\rho_{\text{partial}} = 0.292$ after controlling for SNR. The heterogeneity observed in real data (Norman $\rho_{\text{partial}} = -0.859$, Dixit $\rho_{\text{partial}} = +0.627$) far exceeds this null prediction, confirming that biological factors beyond simple SNR confounding drive the magnitude–coherence relationship.

PCA vs scGPT coherence concordance



Supplementary Figure S7: **Cross-representation concordance of Shesha_p between the frozen PCA representation and the scGPT embedding.** Each point is one perturbation, plotted as coherence computed in frozen PCA against coherence recomputed in scGPT. Dashed line is identity. Spearman ρ and n annotated per panel. Concordance is moderate (0.471–0.829) and points fall predominantly below identity, so scGPT coherence is systematically lower than PCA coherence. This is distinct from the within-scGPT redundancy check in Fig. 2, where both axes are scGPT quantities. Papalexi ($n = 24$) and the Adamson pilot ($n = 8$) are omitted: coherence rank statistics are not numerically reproducible under GPU inference at small n (Sec. 3.6).



Supplementary Figure S8: **The coherence–magnitude relationship holds across effect-size definitions and representations.** Spearman ρ between Shesha_{ap} and effect magnitude computed three ways: centroid displacement in frozen PCA, energy distance, and both quantities recomputed inside the scGPT embedding. Papalexi and the Adamson pilot are omitted from the scGPT bars.

Table S1: Dataset overview for the frozen analysis cohort. Frozen cells include retained perturbed and control cells after filtering for at least 50 cells per perturbation and hash-stable downsampling to at most 100 cells per perturbation and 5,000 controls (seed 320). These are analysis-cohort counts, not raw source dataset totals.

Dataset	Modality	Cell type	Scored perturbations	Frozen cells	Median cells/pert.	Control label
Norman 2019 (Norman et al. 2019)	CRISPRa	K562	236	28,290	100	control
Adamson 2016 UPR (Adamson et al. 2016)	CRISPRi	K562	87	13,605	100	control
Adamson 2016 pilot (Adamson et al. 2016)	CRISPRi	K562	8	810	100	control
Dixit 2016 (Dixit et al. 2016)	CRISPR-KO	BMDC	98	13,664	100	control
Papalexi 2021 (Papalexi et al. 2021)	CRISPR-KO	THP-1	24	4,786	100	NT
Replogle 2022 (Replogle et al. 2022)	CRISPRi	K562	1,832	175,013	100	control

Table S2: Pipeline supersession. Real-data manuscript tables use a single frozen stamp (pipeline_config.CONFIG_VERSION). 2026-08-04.1 is the same freeze as 2026-07-29.1 under a calendar label (CONFIG_VERSION_ALIASES); do not treat the two names as different digests. Synthetic and efficiency-filter lanes are versioned separately and must not be mixed with freeze numbers.

Label	Scope	Status
2026-07-29.1	All real-data tables	Current. Hash-stable downsample; seed 320
2026-08-04.1	Calendar alias of 2026-07-29.1	Same freeze and digest. Appears in early Drive/TRADE stamps
2026-07-25.1	matrix_is_log + positional downsample	Retired. Pathway mag drifted at the same stamp
2026-08-03.1	Calendar alias of 2026-07-25.1	Retired with that stamp
2026-07-17.5	Heuristic Adamson skip-normalize	Retired
SYNTHETIC 2026-07-25.9	scDesign3 generator only	Independent of freeze. manuscript_usable=False
EFFICIENCY_FILTER_VERSION	PS / Mixscape sensitivity	Does not replace frozen_sp_scores.csv

See methods_config_supersession.md and pipeline_config.CONFIG_VERSION_ALIASES.

Table S3: Per-dataset `matrix_is_log` pin from `.X` inspection (`inspect_matrix_scale.py`). Integer-like / large-max matrices are raw counts (`False` $\rightarrow$ normalize + `log1p`). Non-integer floats with max ~ 5 – 10 are already log (`True` $\rightarrow$ skip). Only Adamson UPR was mis-pinned under the heuristic era.

Dataset	<code>max(.X)</code>	frac near-int	pin	verified
Adamson pilot	47.000	—	False	True
Pins live in <code>DATASETS[*][matrix_is_log]</code> .				

Table S4: Reproducibility verification under CONFIG_VERSION=2026-07-29.1. The load-bearing claim is bit-identical materialize and Sp across all six datasets. GPU scGPT ranks at $n = 24$ are the documented exception and are omitted from the scGPT series.

Check	Scope	Result	Detail
Materialize + Sp	six datasets, $n = 2,285$	bit-identical	$\max dS_p = 0$ (verified twice)
Digest guard	<code>assert_frozen_sp_compatible</code>	aborts	four tested mismatch paths
Bootstrap CIs	$n = 10,000$, seed 320	bit-identical	-verify-bootstrap
Papalexi GEO vs freeze	$n = 24$ shared genes	$\rho = 1.000$	independent GSE153056 rebuild
Papalexi scGPT (GPU)	$n = 24$, three runs	not reproducible	Sp vs freeze 0.575–0.633; mag $\rho = 0.753913$ bit-identical
Cross-seed frozen-linear Sp	Norman / Replogle, 15 seeds	$r = 1$	analytic path; see S8

Table S5: Linear slope of coherence on effect magnitude, pooled across datasets. Ordinary least squares fit of within-dataset z-scored Sheshap on within-dataset z-scored effect magnitude, with dataset fixed effects. A unit increase in standardized magnitude corresponds to a 0.956 standard-deviation increase in coherence (SE = 0.006), giving a scale-free linear companion to the pooled rank correlation in Fig. 2g. Because both axes are z-scored within dataset before fitting, the intercept and the dataset fixed effects are zero by construction and are not interpretable. This fit estimates the pooled slope only: it does not decompose variance across datasets and does not include the spread or cell-count terms described in Methods.

Term	coef	SE	Model
Intercept	-0.0000	0.0316	ols_dataset_fe_fallback
mag_z	0.9557	0.0062	ols_dataset_fe_fallback

Table S6: Coherence–magnitude correlation across effect-size definitions. Spearman ρ between Sheshap and four measures of perturbation effect size, all computed in the frozen 50-PC embedding: centroid Euclidean displacement, which is the measure used throughout the manuscript; energy distance, which additionally penalizes within-group dispersion; centroid displacement in a whitened space; and mean distance to k -nearest matched control cells. The relationship is not specific to how effect size is defined, with correlations of 0.843 or above under every measure in every dataset. The largest difference between measures is in Dixit, where energy distance and the whitened metric give substantially higher values than centroid displacement (0.945 and 0.965 against 0.843), consistent with a large share of that dataset’s residual under centroid displacement being within-group dispersion (Fig. 3). Values from the pre-freeze pipeline are not carried over.

Dataset	n	$\rho(S_p, M_p)$	$\rho(S_p, Ed)$	ρ_{Whitened}	$\rho_{k\text{-NN}}$
Norman	236	0.950	0.960	0.963	0.951
Adamson UPR	87	0.951	0.971	0.976	0.929
Adamson pilot	8	0.976	0.976	0.976	0.929
Dixit	98	0.843	0.945	0.965	0.949
Papalexi	24	0.945	0.955	0.988	0.988
Repogle	1,832	0.977	0.980	0.983	0.978

Euclidean / centroid magnitude and energy distance from the frozen 50-PC embedding. Pre-freeze Dixit 0.746 / Papalexi 0.981 are not carried over.

Table S7: Linear-embedding dimensionality ablation. Magnitude–Shesha Spearman ρ with 95% bootstrap CIs (10,000 iterations, seed 320) across embedding dimensions under 2026-07-29.1. The 50-dimensional row is a live reproduction of Table 1, not an imported value. Every dimension refits the recorded dataset-specific backend (scanpy PCA or sparse TruncatedSVD) on the freeze cell set. Adamson UPR ($n = 87$) and Papalexi ($n = 24$) belong in this table when the freeze sweep includes them.

Dataset	Dimensions	ρ	95% CI	p
Adamson UPR	10	0.942	[+0.895, +0.967]	3.6×10^{-42}
Adamson UPR	20	0.951	[+0.909, +0.972]	4.2×10^{-45}
Adamson UPR	30	0.953	[+0.912, +0.974]	6.5×10^{-46}
Adamson UPR	50	0.951	[+0.909, +0.972]	3.2×10^{-45}
Adamson UPR	100	0.955	[+0.915, +0.975]	7.8×10^{-47}
Dixit	10	0.812	[+0.702, +0.888]	3.5×10^{-24}
Dixit	20	0.820	[+0.718, +0.891]	5.2×10^{-25}
Dixit	30	0.839	[+0.744, +0.901]	4.3×10^{-27}
Dixit	50	0.843	[+0.753, +0.902]	1.4×10^{-27}
Dixit	100	0.827	[+0.732, +0.892]	1.1×10^{-25}
Norman	10	0.948	[+0.928, +0.959]	4.0×10^{-118}
Norman	20	0.951	[+0.933, +0.961]	3.2×10^{-121}
Norman	30	0.948	[+0.930, +0.960]	1.3×10^{-118}
Norman	50	0.950	[+0.930, +0.961]	8.0×10^{-120}
Norman	100	0.946	[+0.925, +0.959]	4.3×10^{-116}
Papalexi	10	0.877	[+0.645, +0.968]	2.0×10^{-8}
Papalexi	20	0.932	[+0.803, +0.981]	3.5×10^{-11}
Papalexi	30	0.937	[+0.806, +0.983]	1.5×10^{-11}
Papalexi	50	0.945	[+0.834, +0.984]	3.6×10^{-12}
Papalexi	100	0.960	[+0.866, +0.990]	1.2×10^{-13}
Repogle	10	0.963	[+0.959, +0.967]	$< 10^{-300}$
Repogle	20	0.971	[+0.968, +0.974]	$< 10^{-300}$
Repogle	30	0.974	[+0.971, +0.977]	$< 10^{-300}$
Repogle	50	0.977	[+0.974, +0.979]	$< 10^{-300}$
Repogle	100	0.980	[+0.977, +0.982]	$< 10^{-300}$

Table S8: Seed and runtime reproducibility. The frozen linear-embedding path is deterministic: 15 analytic seeds give identical Spearman ρ (cross-seed $r = 1$, not a rounded 0.99997). The interesting exception is Papalexi scGPT under GPU inference at $n = 24$: Shesha ranks move across runs while magnitude concordance is bit-identical. That dataset is omitted from every scGPT series (same thin- n rule as the Adamson pilot, $n = 8$).

Dataset	Quantity	Runs	Mean / span	Other	Verdict
Norman / Replogle	Frozen linear Sp	15	1.000000 ± 0.000000	1.000000	analytic; not the 0.99997 GPU-era figure exclude from scGPT series
Papalexi scGPT (GPU)	vs frozen Sp	3	0.613 (span 0.575–0.633)	0.753913043	

Table S9: Leave-one-out influence on the magnitude–Shesha Spearman correlation. $\Delta\rho$ is ρ after dropping one perturbation minus the full-sample ρ . Dixit is $n = 98$ under the freeze (the prior version $n = 153$ used MIN_CELLS= 10). No single perturbation drives the relationship in any scoreable dataset.

Dataset	n	Full ρ	LOO range	max $ \Delta\rho $	Largest $+\Delta\rho$	Largest $-\Delta\rho$
Adamson UPR	87	0.951	[0.950, 0.957]	0.0060	GBF1 (+0.0060)	CARS (-0.0017)
Adamson pilot	8	0.976	[0.964, 1.000]	0.0238	CREB1_pDS269 (+0.0238)	62(mod)_pBA581 (-0.0119)
Dixit	98	0.843	[0.838, 0.856]	0.0127	INTERGENIC216151+EGR1 (+0.0127)	CREB1 (-0.0049)
Norman	236	0.950	[0.949, 0.952]	0.0024	CEBPA+KLF1 (+0.0024)	TGFBR2 (-0.0006)
Papalexi	24	0.945	[0.938, 0.960]	0.0153	CD86 (+0.0153)	CAV1 (-0.0075)
Repogle	1,832	0.977	[0.977, 0.977]	0.0004	CHMP2A (+0.0004)	EIF2S1 (-0.0000)

Table S10: Rank-based partial Spearman between Shesha and effect magnitude, controlling for intrinsic spread and per-perturbation cell count. This is an SNR diagnostic, not a headline claim. The paper no longer argues from discordance rankings, a UPR association, or reproducibility prediction; sign heterogeneity across datasets is therefore reported, not explained away. Method: `stats_utils.partial_spearman_rank` (pingouin method='spearman', or the equivalent rank→residualize→Pearson fallback).

Dataset	ρ_{partial}	95% CI	p
Norman	-0.859	[-0.905, -0.781]	6.7×10^{-70}
Dixit 2016	+0.627	[+0.482, +0.728]	4.0×10^{-18}
Repogle	-0.789	[-0.812, -0.765]	$< 10^{-300}$
Pooled	-0.102	[-0.156, -0.049]	1.5×10^{-6}

Table S11: Partial-correlation arithmetic. Several scripts in an earlier version of this analysis residualized on the raw scale and then took Spearman correlations of the residuals. That is not equivalent to a rank-based partial correlation when the relationship between the conditioning variable and the outcome is nonlinear, which it is here: coherence saturates with effect magnitude (Fig. S1). The manuscript default is therefore the rank-based partial Spearman, computed as three steps below (`pingouin.partial_corr(..., method='spearman')`, with a numpy fallback in `stats_utils._partial_spearman_rank_numpy`). Bootstrap confidence intervals resample perturbations and repeat all three steps (10,000 resamples, seed 320). All pathway and stress-marker survival verdicts use this estimator.

Step	Operation
1	Rank x , y , and each covariate in Z
2	Residualize $\text{rank}(x)$ and $\text{rank}(y)$ on $[1, \text{rank}(Z)]$ by OLS
3	Pearson correlation of those residuals = partial Spearman ρ
Legacy (retired)	Residualize raw x, y on raw Z , then Spearman. Not equivalent when the magnitude-Shesha map is nonlinear.
See <code>stats_utils.py</code> . Survival of pathway / stress rows uses this estimator, not the legacy residual Spearman.	

Table S12: Single-marker stress associations with Sheshap. Five scoreable datasets (Adamson UPR $n = 87$; the $n = 8$ pilot is excluded throughout) and four covariate models: magnitude; centroid + QC; E-distance + QC; joint (centroid + E-distance + QC). Survival follows `ci_and_fdr.v1` and applies to the three QC-conditioned models within each dataset; it does not imply a common direction across datasets. Magnitude-only rows are reported for comparison and are not passed through the gate (Status $-$). Papalexi magnitude-only intervals are shown (10,000 valid bootstrap resamples), whereas its three QC-conditioned intervals are omitted because the $n = 24$ QC design is rank-deficient and yielded zero valid resamples. $\star$, shown on the joint row, marks marker–dataset pairs surviving all three QC-conditioned models (Replogle ATF4, DDIT3 and XBP1). ATF4 illustrates why single markers are unreliable here: it survives positively in Adamson UPR (centroid+QC $\rho = +0.340$; E-distance+QC $+0.315$) and negatively in Replogle (-0.106 , -0.106 , joint -0.111), with both screens CRISPRi in K562. HSPA5 survives nowhere under the QC-conditioned models.

Marker	Dataset	n	Model	ρ	95% CI	Status
DDIT3	Norman	236	magnitude	-0.237	[-0.351, -0.107]	survives
	Norman	236	centroid+QC	-0.136	[-0.261, +0.001]	does_not_survive
	Norman	236	edistance+QC	-0.145	[-0.259, -0.007]	indeterminate
	Norman	236	centroid+edistance+QC	-0.143	[-0.265, -0.005]	indeterminate
	Adamson UPR	87	magnitude	-0.067	[-0.237, +0.120]	does_not_survive
	Adamson UPR	87	centroid+QC	-0.036	[-0.224, +0.153]	does_not_survive
	Adamson UPR	87	edistance+QC	-0.070	[-0.253, +0.129]	does_not_survive
	Adamson UPR	87	centroid+edistance+QC	-0.131	[-0.303, +0.113]	does_not_survive
	Dixit	98	magnitude	-0.314	[-0.498, -0.112]	survives
	Dixit	98	centroid+QC	-0.353	[-0.539, -0.140]	survives
	Dixit	98	edistance+QC	-0.194	[-0.436, +0.072]	does_not_survive
	Dixit	98	centroid+edistance+QC	-0.132	[-0.410, +0.151]	does_not_survive
	Papalexi	24	magnitude	-0.248	[-0.584, +0.142]	does_not_survive
	Papalexi	24	centroid+QC	-0.235	—	descriptive_small_n
	Papalexi	24	edistance+QC	-0.349	—	descriptive_small_n
	Papalexi	24	centroid+edistance+QC	-0.394	—	descriptive_small_n
	Replogle	1,832	magnitude	-0.250	[-0.288, -0.208]	survives
	Replogle	1,832	centroid+QC	-0.297	[-0.337, -0.253]	survives
	Replogle	1,832	edistance+QC	-0.287	[-0.328, -0.239]	survives
	Replogle	1,832	centroid+edistance+QC	-0.303	[-0.343, -0.260]	survives $\star$
ATF4	Norman	236	magnitude	+0.395	[+0.275, +0.487]	survives
	Norman	236	centroid+QC	+0.140	[+0.008, +0.258]	indeterminate
	Norman	236	edistance+QC	+0.146	[+0.016, +0.262]	indeterminate
	Norman	236	centroid+edistance+QC	+0.123	[-0.007, +0.251]	does_not_survive
	Adamson UPR	87	magnitude	+0.250	[-0.018, +0.473]	does_not_survive
	Adamson UPR	87	centroid+QC	+0.340	[+0.052, +0.534]	survives
	Adamson UPR	87	edistance+QC	+0.315	[+0.040, +0.512]	survives
	Adamson UPR	87	centroid+edistance+QC	+0.150	[-0.088, +0.436]	does_not_survive
	Dixit	98	magnitude	-0.176	[-0.356, +0.009]	does_not_survive
	Dixit	98	centroid+QC	-0.244	[-0.438, -0.032]	survives
	Dixit	98	edistance+QC	-0.218	[-0.424, +0.007]	does_not_survive
	Dixit	98	centroid+edistance+QC	-0.305	[-0.483, -0.065]	survives
	Papalexi	24	magnitude	+0.009	[-0.431, +0.497]	does_not_survive
	Papalexi	24	centroid+QC	-0.152	—	descriptive_small_n
	Papalexi	24	edistance+QC	-0.259	—	descriptive_small_n
	Papalexi	24	centroid+edistance+QC	-0.279	—	descriptive_small_n
	Replogle	1,832	magnitude	-0.136	[-0.175, -0.096]	survives
	Replogle	1,832	centroid+QC	-0.106	[-0.148, -0.062]	survives
	Replogle	1,832	edistance+QC	-0.106	[-0.148, -0.063]	survives
	Replogle	1,832	centroid+edistance+QC	-0.111	[-0.152, -0.068]	survives $\star$

Table S12 Continued

Marker	Dataset	<i>n</i>	Model	ρ	95% CI	Status
XBP1	Norman	236	magnitude	+0.287	[+0.159, +0.396]	survives
	Norman	236	centroid+QC	+0.089	[-0.044, +0.212]	does_not_survive
	Norman	236	edistance+QC	+0.062	[-0.071, +0.187]	does_not_survive
	Norman	236	centroid+edistance+QC	+0.009	[-0.113, +0.156]	does_not_survive
	Adamson UPR	87	magnitude	+0.123	[-0.087, +0.327]	does_not_survive
	Adamson UPR	87	centroid+QC	+0.251	[-0.009, +0.454]	does_not_survive
	Adamson UPR	87	edistance+QC	+0.209	[-0.034, +0.398]	does_not_survive
	Adamson UPR	87	centroid+edistance+QC	+0.112	[-0.097, +0.368]	does_not_survive
	Dixit	98	magnitude	-0.139	[-0.328, +0.064]	does_not_survive
	Dixit	98	centroid+QC	-0.143	[-0.341, +0.075]	does_not_survive
	Dixit	98	edistance+QC	-0.110	[-0.323, +0.125]	does_not_survive
	Dixit	98	centroid+edistance+QC	-0.067	[-0.291, +0.167]	does_not_survive
	Papalexi	24	magnitude	-0.163	[-0.527, +0.370]	does_not_survive
	Papalexi	24	centroid+QC	-0.256	—	descriptive_small_n
	Papalexi	24	edistance+QC	-0.373	—	descriptive_small_n
	Papalexi	24	centroid+edistance+QC	-0.394	—	descriptive_small_n
	Replogle	1,832	magnitude	-0.239	[-0.279, -0.198]	survives
	Replogle	1,832	centroid+QC	-0.196	[-0.239, -0.153]	survives
	Replogle	1,832	edistance+QC	-0.199	[-0.240, -0.155]	survives
	Replogle	1,832	centroid+edistance+QC	-0.201	[-0.242, -0.158]	survives*
HSPA5	Norman	236	magnitude	-0.036	[-0.170, +0.104]	does_not_survive
	Norman	236	centroid+QC	-0.125	[-0.255, +0.020]	does_not_survive
	Norman	236	edistance+QC	-0.151	[-0.276, -0.007]	indeterminate
	Norman	236	centroid+edistance+QC	-0.166	[-0.285, -0.010]	indeterminate
	Adamson UPR	87	magnitude	+0.048	[-0.140, +0.244]	does_not_survive
	Adamson UPR	87	centroid+QC	+0.116	[-0.098, +0.315]	does_not_survive
	Adamson UPR	87	edistance+QC	+0.056	[-0.153, +0.260]	does_not_survive
	Adamson UPR	87	centroid+edistance+QC	-0.055	[-0.223, +0.217]	does_not_survive
	Dixit	98	magnitude	-0.141	[-0.345, +0.067]	does_not_survive
	Dixit	98	centroid+QC	-0.083	[-0.292, +0.120]	does_not_survive
	Dixit	98	edistance+QC	+0.011	[-0.230, +0.221]	does_not_survive
	Dixit	98	centroid+edistance+QC	+0.001	[-0.246, +0.210]	does_not_survive
	Papalexi	24	magnitude	-0.152	[-0.656, +0.394]	does_not_survive
	Papalexi	24	centroid+QC	-0.205	—	descriptive_small_n
	Papalexi	24	edistance+QC	-0.252	—	descriptive_small_n
	Papalexi	24	centroid+edistance+QC	-0.264	—	descriptive_small_n
	Replogle	1,832	magnitude	-0.128	[-0.172, -0.083]	survives
	Replogle	1,832	centroid+QC	-0.094	[-0.139, -0.049]	does_not_survive
	Replogle	1,832	edistance+QC	-0.074	[-0.120, -0.028]	does_not_survive
	Replogle	1,832	centroid+edistance+QC	-0.078	[-0.123, -0.033]	does_not_survive

Table S13: Pathway-level partial correlations with Sheshap. Rank-partial Spearman between coherence and Hallmark Apoptosis and p53 scores, for every dataset–pathway pair under four covariate models ($5 \times 2 \times 4 = 40$ rows). The three quality-controlled models condition on effect size together with per-perturbation mitochondrial fraction, detected genes, and total counts; the magnitude-only rows condition on effect size alone and are reported for comparison rather than passed through the survival gate (Status —). Adamson UPR is included at $n = 87$; the $n = 8$ pilot is excluded throughout. Status follows the gate defined in Methods: an association is marked as surviving when the interval excludes zero, $|\rho| > 0.1$, and BH-FDR < 0.05 , with Benjamini–Hochberg applied across the nine scored features within each dataset and model. Norman p53 is indeterminate because its nearer interval bound falls inside the knife-edge margin ($\epsilon = 0.02$). Papalexi ($n = 24$) is point-estimate only, as its quality-covariate models are rank-deficient in bootstrap resamples. Hallmark UPR is not among the manuscript pathways and is not shown.

Pathway	Dataset	n	Model	ρ	95% CI	p	Status
Apoptosis	Norman	236	magnitude	-0.349	—	3.8×10^{-8}	—
Apoptosis	Norman	236	centroid+QC	-0.378	[-0.503, -0.211]	2.7×10^{-9}	survives
Apoptosis	Norman	236	edistance+QC	-0.376	[-0.492, -0.211]	2.4×10^{-9}	survives
Apoptosis	Norman	236	centroid+edistance+QC	-0.329	[-0.459, -0.176]	2.2×10^{-7}	survives
Apoptosis	Adamson UPR	87	magnitude	-0.299	—	0.005	—
Apoptosis	Adamson UPR	87	centroid+QC	-0.356	[-0.534, -0.134]	9.7×10^{-4}	survives
Apoptosis	Adamson UPR	87	edistance+QC	-0.308	[-0.495, -0.073]	0.004	survives
Apoptosis	Adamson UPR	87	centroid+edistance+QC	-0.141	[-0.416, +0.053]	0.192	does_not_survive
Apoptosis	Dixit	98	magnitude	-0.057	—	0.577	—
Apoptosis	Dixit	98	centroid+QC	-0.133	[-0.325, +0.088]	0.202	does_not_survive
Apoptosis	Dixit	98	edistance+QC	-0.146	[-0.343, +0.088]	0.151	does_not_survive
Apoptosis	Dixit	98	centroid+edistance+QC	-0.154	[-0.382, +0.099]	0.131	does_not_survive
Apoptosis	Papalexi	24	magnitude	-0.447	—	0.032	—
Apoptosis	Papalexi	24	centroid+QC	-0.363	—	0.097	descriptive_small_n
Apoptosis	Papalexi	24	edistance+QC	-0.232	—	0.275	descriptive_small_n
Apoptosis	Papalexi	24	centroid+edistance+QC	-0.221	—	0.299	descriptive_small_n
Apoptosis	Replogle	1,832	magnitude	-0.294	—	6.1×10^{-38}	—
Apoptosis	Replogle	1,832	centroid+QC	-0.291	[-0.329, -0.248]	5.6×10^{-37}	survives
Apoptosis	Replogle	1,832	edistance+QC	-0.274	[-0.313, -0.231]	7.3×10^{-33}	survives
Apoptosis	Replogle	1,832	centroid+edistance+QC	-0.289	[-0.328, -0.248]	1.5×10^{-36}	survives
p53	Norman	236	magnitude	-0.333	—	1.7×10^{-7}	—
p53	Norman	236	centroid+QC	-0.152	[-0.274, -0.015]	0.020	indeterminate
p53	Norman	236	edistance+QC	-0.147	[-0.262, -0.010]	0.024	indeterminate
p53	Norman	236	centroid+edistance+QC	-0.112	[-0.237, +0.014]	0.086	does_not_survive
p53	Adamson UPR	87	magnitude	-0.063	—	0.565	—
p53	Adamson UPR	87	centroid+QC	-0.071	[-0.272, +0.147]	0.524	does_not_survive
p53	Adamson UPR	87	edistance+QC	-0.102	[-0.296, +0.123]	0.345	does_not_survive
p53	Adamson UPR	87	centroid+edistance+QC	-0.124	[-0.315, +0.121]	0.254	does_not_survive
p53	Dixit	98	magnitude	-0.247	—	0.015	—
p53	Dixit	98	centroid+QC	-0.331	[-0.490, -0.140]	0.001	survives
p53	Dixit	98	edistance+QC	-0.269	[-0.451, -0.047]	0.007	survives
p53	Dixit	98	centroid+edistance+QC	-0.241	[-0.452, -0.004]	0.017	indeterminate
p53	Papalexi	24	magnitude	-0.398	—	0.060	—
p53	Papalexi	24	centroid+QC	-0.276	—	0.214	descriptive_small_n
p53	Papalexi	24	edistance+QC	-0.206	—	0.335	descriptive_small_n
p53	Papalexi	24	centroid+edistance+QC	-0.196	—	0.360	descriptive_small_n
p53	Replogle	1,832	magnitude	-0.400	—	2.1×10^{-71}	—
p53	Replogle	1,832	centroid+QC	-0.382	[-0.415, -0.343]	1.7×10^{-64}	survives
p53	Replogle	1,832	edistance+QC	-0.356	[-0.396, -0.311]	5.1×10^{-56}	survives
p53	Replogle	1,832	centroid+edistance+QC	-0.379	[-0.413, -0.341]	1.1×10^{-63}	survives

Table S14: The apoptosis and p53 associations are not one result reported twice. Hallmark Apoptosis and Hallmark p53 Pathway share 11 of 113 union genes (Jaccard 0.097), and their composite scores correlate weakly to moderately in every dataset contributing to the survival gate ($\rho = 0.145$ to 0.423). Papalexi ($\rho = 0.723$) is excluded from the gate because its quality-covariate models are rank-deficient at $n = 24$, and it contributes to neither pathway result. Dixit is the clearest case for independence: the two scores correlate at 0.423 , yet the partial correlations diverge, with p53 surviving conditioning and apoptosis null. The gene-set Jaccard is a property of the Hallmark sets and is identical across datasets; the score correlation is dataset-specific and is the informative column.

Dataset	n	score ρ	Jaccard (full)	shared/union	gate-relevant
Norman	236	0.153	0.097	11/113	true
Adamson UPR	87	0.145	0.097	11/113	true
Dixit	98	0.423	0.097	11/113	true
Papalexi	24	0.723	0.097	11/113	false
Replogle	1,832	0.295	0.097	11/113	true

Table S15: E-distance competitor on the frozen 50-PC embedding. ρ is Spearman; residual fractions are $1 - R^2$ from rank-OLS of $\text{rank}(S_p)$ on $\text{rank}(\text{size})$. CIs use 2,000 bootstrap replicates, seed 320. Dixit’s residual drop $0.289 \rightarrow 0.108$ is the dispersion correction.

Dataset	n	$\rho(M_p, \text{Ed})$	$\rho(S_p, M_p)$	$\rho(S_p, \text{Ed})$	frac cent.	frac Ed
Norman	236	0.998	0.950	0.960	0.098	0.078
Adamson UPR	87	0.995	0.951	0.971	0.095	0.058
Adamson pilot	8	1.000	0.976	0.976	0.047	0.047
Dixit	98	0.950	0.843	0.945	0.289	0.108
Papalexi	24	0.991	0.945	0.955	0.107	0.088
Repogle	1,832	0.994	0.977	0.980	0.046	0.039

Table S16: TRADE transcriptome-wide impact (TWI) competitor (R ashv half-uniform mixture). The informative contrast is $\rho(S_p, \text{TWI})$ vs $\rho(M_p, \text{TWI})$, not $\rho(S_p, \text{TWI})$ vs $\rho(S_p, M_p)$. Pilot $n = 8$ is uninformative.

Dataset	n	$\rho(S_p, M_p)$	$\rho(S_p, \text{TWI})$	$\rho(M_p, \text{TWI})$	$\rho(S_p, \text{Ed})$
Norman	236	0.950	0.688	0.812	0.960
Adamson UPR	87	0.951	0.954	0.968	0.971
Adamson pilot	8	0.976	0.857	0.833	0.976
Dixit	98	0.843	0.452	0.513	0.945
Papalexi	24	0.945	0.858	0.832	0.955
Replogle	1,832	0.977	0.912	0.919	0.980

Table S17: TWI versus Hallmark pathway-score diagnostic. Checks whether TRADE's transcriptome-wide impact merely restates the apoptosis / p53 scores used in Approach A.

Dataset	Pathway	n	$\rho(\text{TWI, score})$	p
Adamson UPR	Apoptosis	87	-0.061	0.576
Adamson UPR	ROS	87	-0.230	0.032
Adamson UPR	UPR	87	+0.335	0.002
Adamson UPR	mTORC1	87	+0.113	0.298
Adamson UPR	p53	87	+0.114	0.294
Adamson pilot	Apoptosis	8	+0.167	0.693
Adamson pilot	ROS	8	-0.643	0.086
Adamson pilot	UPR	8	-0.024	0.955
Adamson pilot	mTORC1	8	-0.881	0.004
Adamson pilot	p53	8	+0.381	0.352
Dixit	Apoptosis	98	-0.218	0.031
Dixit	ROS	98	+0.096	0.345
Dixit	UPR	98	-0.157	0.124
Dixit	mTORC1	98	-0.351	3.9×10^{-4}
Dixit	p53	98	-0.403	4.0×10^{-5}
Norman	Apoptosis	236	-0.002	0.970
Norman	ROS	236	-0.391	4.6×10^{-10}
Norman	UPR	236	-0.085	0.192
Norman	mTORC1	236	-0.780	1.4×10^{-49}
Norman	p53	236	+0.542	1.9×10^{-19}
Papalexi	Apoptosis	24	-0.458	0.024
Papalexi	ROS	24	+0.009	0.968
Papalexi	UPR	24	-0.254	0.231
Papalexi	mTORC1	24	+0.047	0.828
Papalexi	p53	24	-0.271	0.200
Replogle	Apoptosis	1,832	+0.203	1.8×10^{-18}
Replogle	ROS	1,832	-0.132	1.3×10^{-8}
Replogle	UPR	1,832	-0.150	1.2×10^{-10}
Replogle	mTORC1	1,832	-0.574	5.9×10^{-161}
Replogle	p53	1,832	+0.260	1.2×10^{-29}

Table S18: Restricting to responder cells leaves the coherence–magnitude relationship unchanged. Spearman ρ between Sheshap and effect magnitude computed on all assigned cells and on cells classified as responders by perturbation scoring, together with the rank correlation between the two sets of perturbation-level coherence values. Changes to the coherence–magnitude correlation span -0.048 to $+0.026$ with no consistent direction, so unperturbed cells in assigned guide pools do not account for the coupling between the two quantities. Rank preservation is classified against pre-specified thresholds (preserved, $\rho \geq 0.9$; change, $\rho < 0.7$); Dixit is the only dataset whose perturbation rankings change materially. Perturbations are dropped when too few responder cells remain, which accounts for the difference between the two sample-size columns and is largest in Replogle (171 of 1,832). Because responder classification selects cells whose displacement aligns with an estimated perturbation signature, coherence itself necessarily rises under this restriction; that increase is not reported as a result. These values are a sensitivity analysis and do not replace the frozen estimates, which use all assigned cells throughout.

Dataset	n all	n resp.	ρ all	ρ resp.	rank ρ	95% CI	Verdict
Norman	236	234	0.950	0.902	0.918	[+0.890, +0.937]	preserved
Adamson UPR	87	80	0.951	0.951	0.865	[+0.789, +0.909]	partial
Dixit	98	96	0.843	0.847	0.642	[+0.502, +0.745]	change
Papalexi	24	24	0.945	0.971	0.785	[+0.516, +0.907]	partial
Replogle	1,832	1,661	0.977	0.976	0.916	[+0.908, +0.923]	preserved

Table S19: Mixscape efficiency-filter arm, Norman only. Repeating the responder-restriction analysis with Mixscape as an independent caller. Mixscape returns no responders for 87 of the 236 perturbations, so the comparison is made on the remaining 149 and both columns are computed on that subset. The all-cell correlation of 0.853 is therefore a property of this subset and is not comparable to the frozen Norman value of 0.950 computed on all 236 perturbations. Within the subset, restricting to responders changes the coherence–magnitude correlation by +0.030, opposite in sign to the −0.048 obtained with perturbation scoring on the same dataset; the two callers agree on only 68.4 percent of individual cells (Table S20). Perturbation rankings are preserved between the full and responder-only analyses within the subset ($\rho = 0.891$).

Dataset	n scored	n resp.	ρ all	ρ resp.	rank ρ	Note
Norman	149	149	0.853	0.883	0.891	149/236 denominator; not a frozen baseline

Table S20: The two standard responder classifiers disagree on roughly half the cells they call. Per-cell responder assignments from perturbation scoring and from Mixscape, compared on the 23,290 Norman cells scored by both. Mixscape is the more conservative caller, and its responder set is close to but not nested within that of perturbation scoring: 2,512 cells are called responders by Mixscape alone. Because which cells count as responders is substantially method-defined, responder-only reanalysis is reported as a sensitivity analysis conditional on the classifier rather than as a correction applied to the frozen estimates, and the direction of the change in the coherence–magnitude correlation differs between the two callers in this dataset (Fig. 4).

Quantity	Value
n shared cells	23,290
PS responder fraction	0.531
Mixscape responder fraction	0.431
Both	7,521
PS only	4,836
Mixscape only	2,512
Jaccard	0.506
Cohen’s κ	0.374
Raw agreement	0.684

Table S21: Coherence does not predict between-guide agreement beyond effect size. Split-half agreement is circular as a measure of reproducibility, because both halves are drawn from the same cells and compared against the same control reference. We therefore used independent guides targeting the same gene, which are separate reagents in separate cells. For each gene with at least two guides passing the cell-count filter, we correlated gene-level Sheshap with the cosine between the two guides' mean response directions, conditioning on effect magnitude. The partial correlation is null in both datasets. In Replogle this endpoint is available for 104 of the 1,832 scored perturbations, so it covers a small and not necessarily representative fraction of the screen. The third row asks a different question: whether guides targeting the same gene give similar coherence values at all. They do ($ICC(1) = 0.890$), but effect magnitude agrees between guides to the same degree ($ICC(1) = 0.889$, difference 0.001), so between-guide consistency in coherence carries no information beyond consistency in effect size. Pairwise Spearman correlations over unordered guide pairs are not reported, because the value depends on which guide of each pair is assigned to x ; $ICC(1)$ is the well-defined concordance statistic for this design. Both datasets share a control reference across guides, so this is independent-reagent rather than fully independent replication.

Dataset	Endpoint	n	Stat	95% CI	Note
Replogle	gene Sp ~ between-guide cosine magnitude	104 genes	0.006	$[-0.220, 0.225]$	null; 104/1,832 of the screen
Papalexi	gene Sp ~ between-guide cosine magnitude	23 genes	0.183	$[-0.317, 0.644]$	null (GEO rebuild)
Papalexi	$ICC(1)$ of guide-level Sp	23 genes	0.890	$[0.752, 0.951]$	$ICC(mag) = 0.889$; $\Delta = 0.001$

Table S22: Papalexi GEO reporting ladder (GSE153056, no pertpy). The prior version $n = 25$ is the gene-level inventory at MIN_CELLS= 10; the freeze uses 50 throughout and yields $n = 24$ — the same cutoff change that took Dixit from 153 to 98. The independent rebuild reproduces frozen gene Sp at $\rho = 1.000$ ($n = 24$).

Rung	Count
Guides deposited (GSE153056)	112
Guides / genes at MIN_CELLS=10 (sensitivity)	96 / 25
Guides / genes at frozen MIN_CELLS=50	83 / 24
GEO vs frozen gene Sp (n shared)	$\rho = 1.000$ ($n = 24$)

Table S23: Within-representation magnitude–Shesha correlations (one of the two quantities formerly mixed in `tab:scgpt_comparison`). Frozen linear is the manuscript Euclidean column (scanpy PCA or sparse TruncatedSVD according to the dataset-specific backend); E-distance and scGPT are the same relationship scored in those representations. scGPT omitted for Papalexi ($n = 24$, GPU ranks not reproducible) and the Adamson pilot ($n = 8$). Diffusion-map / PHATE within-arm rows appear when those CSVs exist; concordance against the frozen linear embedding is Table S24.

Dataset	n	Frozen linear $\rho(S_p, M_p)$	E-distance ρ	scGPT ρ
Norman	236	0.950	0.960	0.935
Adamson UPR	87	0.951	0.971	0.912
Adamson pilot	8	0.976	0.976	omit
Dixit	98	0.843	0.945	0.854
Papalexi	24	0.945	0.955	omit
Replegle	1,832	0.977	0.980	0.851

Table S24: Three-arm embedding concordance against the frozen linear embedding (scanpy PCA or sparse TruncatedSVD, according to the frozen dataset-specific backend). Left block: Sheshap computed in each alternative embedding against frozen Sheshap. Right block: effect magnitude computed in each embedding against frozen effect magnitude. Diffusion maps are the only representation in which coherence transfers as well as or better than magnitude in every dataset; under PHATE and scGPT magnitude transfers better throughout, and the PHATE gap is widest where coherence is weakest (Replogle 0.44 vs 0.68). MDS convergence does not account for the PHATE pattern (Table S25). scGPT is omitted for Papalexi ($n = 24$) and the Adamson pilot ($n = 8$): coherence rank statistics are not numerically reproducible under GPU inference at those sample sizes.

Dataset	n	DM S_p	PHATE S_p	scGPT S_p	DM M_p	PHATE M_p	scGPT M_p
Norman	236	0.858	0.882	0.514	0.886	0.891	0.629
Adamson UPR	87	0.948	0.705	0.829	0.949	0.775	0.873
Adamson pilot	8	0.690	0.667	omit	0.476	0.762	omit
Dixit	98	0.641	0.684	0.471	0.538	0.794	0.549
Papalexi	24	0.852	0.506	omit	0.808	0.844	omit
Replogle	1,832	0.920	0.439	0.750	0.902	0.679	0.784

Table S25: PHATE convergence control. MDS solver convergence does not predict concordance with the frozen embedding: the three datasets whose embeddings converged span $\rho = 0.51$ to 0.68 , while the three that did not span 0.44 to 0.88 . Norman failed to converge and has the highest PHATE concordance in the set; Papalexi converged and has one of the lowest. The PHATE degradation in Figure 4 is therefore a property of the embedding rather than optimizer failure. No rows are excluded.

Dataset	n	Solver	Converged	ρ vs frozen S_p
Adamson pilot	8	SGD-MDS	yes	0.667
Dixit	98	SGD-MDS	yes	0.684
Papalexi	24	SGD-MDS	yes	0.506
Norman	236	SGD-MDS	no (-3.5%)	0.882
Adamson UPR	87	SGD-MDS	no (-4.0%)	0.705
Replogle	1,832	SGD-MDS	no (-5.1%)	0.439

Table S26: Approach B covariate balance before and after caliper matching on residual Shesha quartiles. Gate: $|\text{SMD}| \leq 0.25$ (inclusive) and ≥ 15 pairs. Magnitude balances in both datasets; cell-quality covariates do not (Replogle n genes -0.46 ; Norman UMIs / genes < -1.2). Replogle n cells after matching is $+0.249$ and therefore passes. Because the gate fails, Approach B is dropped and Approach A carries the pathway result.

Dataset	Stage	Covariate	n pairs	SMD	Gate
Replogle	before	magnitude	458	+0.084	pass
Replogle	before	n_cells	458	+0.266	fail
Replogle	before	n_counts	458	-0.178	pass
Replogle	before	n_genes	458	-0.489	fail
Replogle	before	percent_mito	458	+0.100	pass
Replogle	after	magnitude	422	+0.013	pass
Replogle	after	n_cells	422	+0.249	pass
Replogle	after	n_counts	422	-0.153	pass
Replogle	after	n_genes	422	-0.460	fail
Replogle	after	percent_mito	422	+0.106	pass
Norman	before	magnitude	59	+0.046	pass
Norman	before	n_cells	59	-0.566	fail
Norman	before	n_counts	59	-1.172	fail
Norman	before	n_genes	59	-1.250	fail
Norman	before	percent_mito	59	+0.141	pass
Norman	after	magnitude	43	-0.024	pass
Norman	after	n_cells	43	-0.533	fail
Norman	after	n_counts	43	-1.248	fail
Norman	after	n_genes	43	-1.342	fail
Norman	after	percent_mito	43	+0.134	pass

Table S27: Calibration of the attempted simulation arm. A perturbation simulator was implemented to test whether coherence distinguishes heterogeneity arising from variable perturbation efficiency from heterogeneity arising from cells engaging different transcriptional programs. It was not usable, for two reasons recorded here. The effect scale needed to reproduce observed coherence depends on which reference perturbation the pipeline selects: at a fixed `effect_scale` of 1.5, simulated coherence was 0.593 against the SLU7 reference and 0.144 against GRPEL1, a fourfold change with no modification to the generator. No portable calibration constant therefore exists, and re-tuning the scale to whichever reference the current freeze selects would be fitting a constant to a known target. Separately, the generator could not hold multi-program response magnitude fixed while varying the number of programs, so the identifiability question the arm was built to answer was not testable as constructed. The comparison of simulated against observed coherence is reported for transparency and is not a pre-specified acceptance criterion: the tolerance was set after the arm had already produced a passing value. Arm tables are retained in the analysis outputs but are not cited in the manuscript.

Item	Value
Simulator configuration	2026-07-25.9
<code>effect_scale</code>	1.5
Simulated coherence, SLU7 reference	0.593
Simulated coherence, GRPEL1 reference	0.144
Observed coherence, GRPEL1	0.397
Efficiency arm, coherence across 16 cells	0.110–0.144
Manuscript status	attempted, not used

The simulation arm is isolated from the real-data pipeline: scoring calls `preprocess(..., matrix_is_log=False)` and `calculate_sp` only, and the real-data freeze remains 2026-07-29.1.

Table S28: Centroid-based distances do not approximate the scMAGeCK perturbation-response score. Spearman correlations in Replogle 2022 between scMAGeCK PS and two per-cell distance measures sometimes used as substitutes for it: mean Euclidean distance from the control centroid, and the same distance whitened by the inverse control covariance. Both correlate weakly with the algorithm-derived score, so a centroid distance should not be treated as a lower-fidelity approximation to PS. Responder classification in this paper uses scMAGeCK PS directly (**Fig. 4**).

Comparison	ρ
Euclidean centroid vs scMAGeCK PS	0.097
Mahalanobis centroid vs scMAGeCK PS	0.149